

Abelian Varieties over Finite Fields

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Introduction

Abelian varieties are higher-dimensional analogues of elliptic curves, as always been taught in a first course in abelian varieties. While elliptic curves can be explicitly given by Weierstrass equations, abelian varieties in general cannot be described by simple, explicit equations, even if they are always finitely presented projective varieties and thus cut out by homogeneous polynomials in some $\mathbb{P}^N$. This requires a more structural approach to their study.

It was W  il who first re-interpreted the theory of abelian varieties in the language of modern algebraic geometry, providing the framework for his proof of Mordell-W  il in higher dimension. This perspective was further developed by John Tate and Jean-Pierre Serre. During a series of seminar talks in the 1960s, they gave some remarkable findings which have fundamentally reshaped our perspectives of abelian varieties. These talks have led to several papers [Tat66] [ST68] [Hon68] [Tat71] [Kat81] that remain influential to date. The aim of this essay is to give an account of these classic content and to see how they motivate topics that people are currently interested in.

Among these results, a central role is played by the **Honda-Tate theorem**. It gives a complete classification of simple abelian varieties over finite fields by the W  il q -numbers which come from the Riemann hypothesis part of W  il conjectures. They are related by the bijection $[A] \mapsto [\pi_A]$ where π_A is the Frobenius. Tate proved the injectivity with his isogeny theorem in 1966. In 1968 Honda showed the surjectivity with his own method. Tate presented this in [Tat71] where he also gave several applications of Honda-Tate.

Another important topic on lifting abelian varieties is **Serre-Tate theorem**, which says that the deformations of abelian varieties are controlled by their p -divisible groups. It has become a standard procedure to pass to p -divisible groups from abelian varieties using Serre-Tate when studying their liftings, because the deformation theory of p -divisible groups is more tractable and can be described in linear-algebraic terms via Dieudonn  -Manin theory.

We start with basic definitions of abelian varieties in Chapter 1. We will not stick to the proofs for most theorems which can be found in any text on abelian varieties such as [Mum70]. Then we introduce abelian varieties with complex multiplication including the definition of CM field and CM type.

In Chapter 2, we prove Tate’s isogeny theorem, which states that to understand the endomorphisms of an abelian variety it suffices to study the G -automorphisms of its Tate modules, which motivates us to study the Galois representations $T_\ell A$. We will use it to analyze the structure of the endomorphism algebra.

Chapter 3 discusses good reductions. We introduce N  ron-Ogg-Shafarevich Criterion which states that an abelian variety over a local field has good reduction if and only if the absolute inertia group acts trivially on its Tate module, and also Shimura-Taniyama formula which examines the behavior of the Frobenius in the CM setting.

The proof of Honda-Tate theorem in Chapter 4 is an overall application of previous chapters. We will show its importance on CM lifting and how it solves a conjecture by Manin on Newton polygon.

In Chapter 5 we shift towards finite flat group schemes and p -divisible groups. We also give an overview of Dieudonn   theory, which classifies finite flat group scheme of p -power order and p -divisible groups by Dieudonn   modules so that linear algebra techniques can be involved.

The last chapter is devoted to the Serre-Tate theorem. We will use it to construct the canonical lifts for ordinary abelian varieties to characteristic zero, and see how we can linearize their deformations with p -divisible groups.

Notations

Throughout this paper, k denotes a perfect field with characteristic p , and A will be an abelian variety.

We write $\Gamma_k = \text{Gal}(\bar{k}/k)$, the absolute Galois group of k .

The endomorphism ring of A is denoted by $\text{End}(A)$, and we set $\text{End}^0(A) = \text{End}(A) \otimes \mathbb{Q}$ to be the endomorphism algebra.

The tangent space of A at identity is denoted by $\text{Tgt}_0 A$.

The ℓ -adic Tate module of A is denoted by $T_\ell(A)$ and we define $V_\ell(A) = T_\ell(A) \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell$.

For $a \in \text{End}^0(A)$, P_a will denote its characteristic polynomial when acting on $V_\ell(A)$ (which is independent of the choice of ℓ)

For an isogeny $f : A \rightarrow B$, we will use $f^\vee$ to denote its dual homomorphism and $\hat{f}$ to denote its dual isogeny.

1 Necessary Background of Abelian Varieties with CM

1.1 Basic definitions

We start with the most generic definition of abelian varieties in algebraic geometry:

Definition 1.1. *An abelian variety over a field K is a smooth connected proper group scheme over K .*

This definition is not really helpful sometimes because it does not reveal many wonderful properties of abelian varieties, including being "abelian".

Proposition 1.1. *Let A/K be an abelian variety. Then A is commutative.*

We need the *Rigidity lemma* to prove this

Lemma 1.1. (*Rigidity lemma*) *Let $f : A \times B \rightarrow C$ be a morphism of varieties over K .*

If A is proper and $f(A \times \{b_0\}) = f(\{a_0\} \times B) = \{c_0\}$ for some $a_0 \in A(K)$, $b_0 \in B(K)$, $c_0 \in C(K)$, then $f(A \times B) = \{c_0\}$.

Proof. See [Mum70] □

Proof of Prop.1.1. It is known two K -morphisms $X \rightarrow Y$ with Y separated are equal if they agree on all closed points of X . So it suffices to check that the image of the map

$$\begin{aligned} A(\overline{K}) \times A(\overline{K}) &\rightarrow A(\overline{K}) \\ (x, y) &\mapsto y \cdot x \cdot y^{-1} \cdot x^{-1} \end{aligned}$$

is the identity e . Notice the restriction of this map to $A(\overline{K}) \times \{e\}$ and to $\{e\} \times A(\overline{K})$ is constantly equal to e . So this follows from the Rigidity lemma. □

It is convenient to think of abelian varieties over $\mathbb{C}$ as complex manifolds where differential geometric techniques come in handy. For example:

Theorem 1.1. (*Analytic uniformization of complex abelian varieties*). *Let A be a g -dimensional abelian variety over $\mathbb{C}$. Then there exists a full-rank lattice $\Lambda \subset \mathbb{C}^g$ such that $A(\mathbb{C})$ is isomorphic (as a Lie group) to $\mathbb{C}^g/\Lambda$. We call such $\mathbb{C}^g/\Lambda$ a **complex torus**.*

Sketch of proof. From the differential geometry of Lie groups, we have the exponential map from $T_{gt_0}A \cong \mathbb{C}^g$ to A itself. One can show that this is a surjective group homomorphism. Its kernel Λ is discrete because the exp map is a local homeomorphism, and it must have full rank since $\mathbb{C}^g/\Lambda \cong A(\mathbb{C})$ is compact. □

Definition 1.2. *An abelian variety over $\mathbb{C}$ is the algebraic variety associated to a complex torus V/Λ where $V \cong \mathbb{C}^g$ such that there exists a positive definite Hermitian form $H : V \times V \rightarrow \mathbb{C}$ with $\text{Im}H(\Lambda \times \Lambda) \subseteq \mathbb{Z}$ (Riemann form). We also call this complex torus algebraizable.*

Remark 1.1. The fact that any complex torus with a Riemann form is an abelian variety over $\mathbb{C}$ as in definition 1.1 follows from the celebrated Kodaira embedding theorem, which essentially says that they can be embedded into projective space.

Definition 1.3. *Let A, B be abelian varieties over K . A morphism $f : A \rightarrow B$ is called an **isogeny** if f is a surjective homomorphism with finite kernel. We say that A is isogenous to B , denoted by $A \sim B$*

Corollary 1.1. *Let $A/\mathbb{C}$ be an abelian variety of dimension g , and $[n]$ be the multiplication by n map. Then $[n]$ is an isogeny, and its kernel $A(\mathbb{C})[n] \cong (\mathbb{Z}/n\mathbb{Z})^g$.*

With the identification $A(\mathbb{C}) \cong \mathbb{C}^g/\Lambda$, the corollary follows easily.

1.2 Line Bundles and Dual Abelian Varieties

This section will address the construction of dual abelian varieties. We have to omit proofs of most theorems, which can be found in [Mum70]

Theorem 1.2. (*Theorem of the Cube*) Let X, Y, Z be geometrically integral k -varieties with X and Y proper. Let $x_0 \in X(k), y_0 \in Y(k), z_0 \in Z(k)$. Suppose a line bundle $\mathcal{L}$ on $X \times Y \times Z$ becomes trivial under the three restrictions

$$\mathcal{L}|_{X \times Y \times \{z_0\}}, \mathcal{L}|_{X \times \{y_0\} \times Z}, \mathcal{L}|_{\{x_0\} \times Y \times Z}.$$

Then $\mathcal{L}$ itself is trivial.

Corollary 1.2. Let A be an abelian variety. Let $\pi_i : A \times A \times A \rightarrow A$ denote the projection map on the i -th factor, and set $\pi_{ij} = \pi_i + \pi_j$ and $\pi_{123} = \pi_1 + \pi_2 + \pi_3$. Let $\mathcal{L}$ be a line bundle on A . Then the line bundle

$$\mathcal{L}' = \pi_{123}^* \mathcal{L} \otimes \pi_{12}^* \mathcal{L}^{-1} \otimes \pi_{23}^* \mathcal{L}^{-1} \otimes \pi_{13}^* \mathcal{L}^{-1} \otimes \pi_1^* \mathcal{L} \otimes \pi_2^* \mathcal{L} \otimes \pi_3^* \mathcal{L}$$

on $A \times A \times A$ is trivial.

Proof. This follows immediately from the theorem of the cube by restricting to $A \times A \times \{0\}$, $A \times \{0\} \times A$ and $\{0\} \times A \times A$ respectively. $\square$

Corollary 1.3. Let A be an abelian variety, and X an arbitrary variety. Let $f, g, h : X \rightarrow A$ be maps of varieties, and $\mathcal{L}$ a line bundle on A . Then the line bundle

$$\mathcal{L}' = (f + g + h)^* \mathcal{L} \otimes (f + g)^* \mathcal{L}^{-1} \otimes (f + h)^* \mathcal{L}^{-1} \otimes (g + h)^* \mathcal{L}^{-1} \otimes f^* \mathcal{L} \otimes g^* \mathcal{L} \otimes h^* \mathcal{L}$$

on X is trivial.

Proof. This follows from Corollary 1.2 by considering the map $X \rightarrow A \times A \times A$ given by (f, g, h) . $\square$

Corollary 1.4. Let $\mathcal{L}$ be a line bundle on an abelian variety A . Then for any $n \in \mathbb{Z}$,

$$[n]^* \mathcal{L} = \mathcal{L}^{n(n+1)/2} \otimes [-1]^* \mathcal{L}^{n(n-1)/2}.$$

In particular, if $\mathcal{L} = [-1]^* \mathcal{L}$ (we say such $\mathcal{L}$ is symmetric), then $[n]^* \mathcal{L} = \mathcal{L}^{n^2}$, and if $\mathcal{L}^{-1} = [-1]^* \mathcal{L}$ (we say such $\mathcal{L}$ is antisymmetric), then $[n]^* \mathcal{L} = \mathcal{L}^n$.

Proof. Use Corollary 1.3 with $f = [n], g = id$ and $h = [-1]$. After rearrangement we get the result by induction. $\square$

Theorem 1.3. (*Theorem of the Square*) Let A be an abelian variety over k and $\mathcal{L}$ a line bundle on A , and $x, y \in A(k)$. Then

$$t_{x+y}^* \mathcal{L} \otimes \mathcal{L} = t_x^* \mathcal{L} \otimes t_y^* \mathcal{L}$$

(Here t_x denotes translation by x .)

Proof. Use Corollary 1.3 with $f = t_x, g = t_y$ and $h = id$. $\square$

Tensor $\mathcal{L}^{-2}$ on both sides and we get $t_{x+y}^* \mathcal{L} \otimes \mathcal{L}^{-1} = (t_x^* \mathcal{L} \otimes \mathcal{L}^{-1}) \otimes (t_y^* \mathcal{L} \otimes \mathcal{L}^{-1})$. This gives a group homomorphism $\phi_{\mathcal{L}} : A(k) \rightarrow Pic(A)$ via $a \mapsto t_a^* \mathcal{L} \otimes \mathcal{L}^{-1}$ for any $\mathcal{L} \in Pic(A)$. Furthermore, we get a group homomorphism $\phi : Pic(A) \rightarrow Hom(A(k), Pic(A))$ via $\mathcal{L} \mapsto \phi_{\mathcal{L}}$. Its kernel consists of all translation invariant line bundles on A , i.e. those $\mathcal{L}$ such that $t_a^* \mathcal{L} = \mathcal{L}$ for any $a \in A(k)$

Definition 1.4. We denote $Pic^0(A)$ to be the subgroup $ker \phi \subseteq Pic(A)$

Definition 1.5. Let $K(\mathcal{L}) = \ker(\phi_{\mathcal{L}}) = \{x \in A(k) : t_x^* \mathcal{L} = \mathcal{L}\}$ be those points that fix $\mathcal{L}$.

$K(\mathcal{L})$ helps to characterize line bundles. One can show that for $\mathcal{L} = \mathcal{O}(D)$ with D an effective divisor, $K(\mathcal{L})$ is finite if and only if $\mathcal{L}$ is ample.

This is the key ingredient in one proof on projectivity of abelian varieties. In fact, 'projective proper group scheme' gives an equivalent definition of abelian varieties. But being projective is a very strong condition compared with others, so the authors are less inclined to include it in the definition.

Corollary 1.5. $[n]$ is an isogeny for an abelian variety A/K .

Proof. Note that for $x \in A[n]$, $[n] \circ t_x = [n]$. Pick an ample line bundle $\mathcal{L}$ on A (we can do so since A is projective). Then $t_x^*([n]^* \mathcal{L}) = [n]^* \mathcal{L}$. This implies $A[n] \subseteq K([n]^* \mathcal{L})$. Corollary 1.4 says $[n]^* \mathcal{L} = \mathcal{L}^{n(n+1)/2} \otimes [-1]^* \mathcal{L}^{n(n-1)/2}$, where both terms on the right side are ample, hence the left is also. Then the above observation gives the finiteness of $A[n]$. Now [dJ, lemma 39.9.8] shows the surjectivity of $[n]$. $\square$

We go back to the map $\phi_{\mathcal{L}} : A(k) \rightarrow \text{Pic}(A)$ via $a \mapsto t_a^* \mathcal{L} \otimes \mathcal{L}^{-1}$. Note that by the theorem of square $t_a^* \mathcal{L} \otimes \mathcal{L}^{-1}$ is translation invariant. So $\phi_{\mathcal{L}}$ actually lands in $\text{Pic}^0(A)$. Moreover, one can show that $K(\mathcal{L})$ is finite (or equivalently, $\mathcal{L}$ is ample), then $\phi_{\mathcal{L}}$ is surjective. Therefore, given an ample line bundle $\mathcal{L}$ on A we now have an isomorphism of groups

$$(A/K(\mathcal{L}))(k) \cong \text{Pic}^0(A)$$

It allows us to put an abelian variety structure on $\text{Pic}^0(A)$.

Definition 1.6. We notate $A^\vee = A/K(\mathcal{L})$, and call this the **dual abelian variety** to A .

Proposition 1.2. (Dual homomorphism) Let $f : A \rightarrow B$ be a homomorphism of abelian varieties over k . There exists an induced homomorphism $f^\vee : B^\vee \rightarrow A^\vee$, called the dual of f .

Concretely, at the level of points, $f^\vee$ is simply the pullback of line bundles from B to A :

$$f^\vee : B^\vee(k) = \text{Pic}^0(B) \rightarrow A^\vee(k) = \text{Pic}^0(A), \quad \mathcal{L} \mapsto f^* \mathcal{L}$$

Theorem 1.4. If $f : A \rightarrow B$ is an isogeny, then $f^\vee : B^\vee \rightarrow A^\vee$ is an isogeny and $\ker(f^\vee) = (\ker f)^\vee$, the Cartier dual of the finite commutative group scheme $\ker f$. In particular, $\deg(f^\vee) = \deg(f)$

Note the dual of $[n]_A$ is just $[n]_{A^\vee}$. So this theorem implies that $A[n]^\vee = A^\vee[n]$

We have seen that $\phi_{\mathcal{L}}$ is an isogeny if $\mathcal{L}$ is ample. In general, A **polarization** of A/k is an isogeny $\lambda : A \rightarrow A^\vee$ such that $\lambda_{\bar{k}} : A(\bar{k}) \rightarrow \text{Pic}^0(A)$ is given by $\lambda_{\bar{k}} = \phi_{\mathcal{L}}$ for some ample line bundle $\mathcal{L}$ on A over $\bar{k}$. The degree of the polarisation λ is its degree as an isogeny. An abelian variety together with a polarisation is called a **polarised abelian variety**.

Definition 1.7. A **principal polarization** is a polarization of degree 1, i.e. an isomorphism.

Remark 1.2. Every abelian variety A/k is $\bar{k}$ -isogenous to a principally polarised abelian variety. We only note this because it is a fundamental result for principally polarized abelian variety.

1.3 Endomorphism Algebras

One main task in the study of abelian varieties is to understand their endomorphism algebras. Because of the following decomposition theorems, most cases can be reduced to simple abelian varieties, whose endomorphism algebras have a relatively clear structure.

Theorem 1.5. (*Poincaré reducibility*) *Let A/k be an abelian variety and let $B \subseteq A$ be an abelian subvariety (i.e. a closed subvariety over k that is closed under the group operations and inherits the structure of an abelian variety). Then there exists another abelian subvariety $B' \subseteq A/k$ such that $B \cap B'$ is a finite set and $B + B' = A$, or equivalently $A \sim B \times B'$.*

Proof. Choose an ample line bundle $\mathcal{L}$ on A . Then the pull back $i^*\mathcal{L} = \mathcal{L}|_B$ is also ample. We have the following commutative diagram

$$\begin{array}{ccc} B & \xrightarrow{i} & A \\ \downarrow \phi_{i^*\mathcal{L}} & & \downarrow \phi_{\mathcal{L}} \\ B^\vee & \xleftarrow{i^\vee} & A^\vee \end{array}$$

Let B' be the connected component of $\phi_{\mathcal{L}}^{-1}(\ker i^\vee)$ in A . Now $z \in B \cap C = B \cap \phi_{\mathcal{L}}^{-1}(\ker i^\vee)$ if and only if $i^\vee \circ \phi_{\mathcal{L}}(z) = t_z^*\mathcal{L} \otimes \mathcal{L}^{-1}|_B$ is trivial if and only if $z \in K(\mathcal{L}|_B)$. This implies that $B \cap C \subseteq K(\mathcal{L}|_B)$ is finite since $\mathcal{L}|_B$ is ample. Surjectivity follows by comparing the dimension. $\square$

Definition 1.8. *An abelian variety A/k is called **simple** if the only abelian subvarieties are $\{e\}$ or A itself. It is called **isotypic** if $A \sim (A')^n$ for some simple abelian variety A'/k and $n \in \mathbb{N}^+$*

Note that the definition of simplicity depends on the base field k . Abelian varieties that are simple over a given field might not be simple after base change to a field extension.

Corollary 1.6. *For any abelian variety A/k , we can write*

$$A \sim \prod_{i=1}^n A_i^{r_i}$$

where each A_i/k is simple, $r_i \in \mathbb{Z} > 0$, and the A_i are pairwise non-isogenous.

Proof. Apply Poincaré reducibility inductively. $\square$

Definition 1.9. *Let $A, B/k$ be abelian varieties. We write $\text{Hom}(A, B)$ for the abelian group of homomorphisms from A to B . We write $\text{Hom}^0(A, B) := \text{Hom}(A, B) \otimes_{\mathbb{Z}} \mathbb{Q}$. Likewise, we write $\text{End}(A) = \text{Hom}(A, A)$, and $\text{End}^0(A) := \text{End}(A) \otimes_{\mathbb{Z}} \mathbb{Q}$ which is called the endomorphism algebra of A .*

Remark 1.3. $\text{Hom}^0(A, B)$ and $\text{End}^0(A)$ depend only on the isogeny classes of A and B , since isogenies become have an inverse in them.

Corollary 1.7. *If A/k is simple, then $\text{End}^0(A)$ is a division algebra. More generally, with A decomposed as in Corollary 1.6, we have*

$$\text{End}^0(A) = \prod_{i=1}^n M_{r_i}(\text{End}^0(A_i))$$

In particular, $\text{End}^0(A)$ is a semisimple $\mathbb{Q}$ -algebra.

Proof. The first part follows from the remark and the fact that for a simple abelian variety, the only endomorphisms are 0 and isogenies. The general decomposition follows from the fact that there are no nontrivial homomorphisms between the non-isogenous simple factors A_i . $\square$

1.4 Rosati Involution and CM field

Definition 1.10. Let A/k be an abelian variety and $f \in \text{End}^0(A)$. Let A be polarized with an isogeny $\phi : A \rightarrow A^\vee$ and ϕ^{-1} be its inverse in $\text{Hom}^0(A^\vee, A)$. The **Rosati involution** is the map $(-)^{\dagger} : \text{End}^0(A) \rightarrow \text{End}^0(A)$, $f \mapsto f^{\dagger} = \phi^{-1} \circ f^{\vee} \circ \phi$, where $f^{\vee}$ is the dual homomorphism of f as in proposition 1.2.

Note that the Rosati involution depends on the polarization we choose. Also, since $[n]^{\dagger} = [n]$, it follows that for $f \in \text{End}^0(A)$ its characteristic polynomial P_f agrees with $P_{f^{\dagger}}$ and hence $\text{Tr}(f) = \text{Tr}(f^{\dagger})$. Moreover,

Proposition 1.3. The Rosati involution is a positive involution, i.e. $\text{Tr}_{\text{End}^0(A)/\mathbb{Q}}(f \cdot f^{\dagger}) > 0$ for all $f \in \text{End}^0(A) \setminus \{0\}$

Proof. See [EvdM, lemma 12.24]. □

Lemma 1.2. For a number fields E , TFAE:

1. There exists a field $E^+ \subset E$ such that $E^+/\mathbb{Q}$ is totally real and E/E^+ is an imaginary quadratic extension.
2. There exists a nontrivial $c \in \text{Aut}(E)$ such that for all embeddings $\tau : E \rightarrow \mathbb{C}$ we have $c' \circ \tau = \tau \circ c$, where c' denotes complex conjugation in $\mathbb{C}$.
3. There exists a field $E^+ \subset E$ with $E^+/\mathbb{Q}$ totally real such that $E = E^+[\alpha]$, where $\alpha^2 \in E^+$ is totally negative, i.e. under every embedding $\tau : E^+ \rightarrow \mathbb{R}$ we have $\tau(\alpha^2) < 0$.

Proof. See [Mil20, theorem 6.1] □

Definition 1.11. A number field E satisfying one of the conditions in Lemma 1.2 is said to be a **CM field**. A **CM algebra** is a finite product of CM fields: $E = E_1 \times \cdots \times E_n$ for CM fields $E_1, \dots, E_n$.

Example 1. Recall that an elliptic curve has complex multiplication if its endomorphism algebra is a quadratic imaginary extension of $\mathbb{Q}$, which is the smallest possible CM field.

Corollary 1.8. A finite composite of CM-subfields of a field is CM; in particular, the Galois closure of a CM-field in any larger field is CM.

Proposition 1.4. Every finite-dimensional $\mathbb{Q}$ -algebra admitting a positive involution is semisimple. Moreover, every finite-dimensional commutative $\mathbb{Q}$ -algebra with positive involution is a product of totally real number fields, with the restricted involution the identity, and CM fields, with the restricted involution the complex conjugation.

Proof. See [Mil20, theorem 1.36 & 1.38] □

1.5 Abelian Varieties with CM

Unlike elliptic curves which are always simple because of dimension, the structure of the endomorphism algebras of abelian varieties could be messy, and this is when non-commutative algebra weapons can be used.

Recall that the **reduced degree** of a semisimple k -algebra B with decomposition $B = \prod B_i$ and B_i simple k -algebras is defined to be

$$[B : k]_{\text{red}} = \sum [B_i : k_i]^{1/2} [k_i : k]$$

where k_i is the center of B_i . In particular, the reduced degree for a central simple k -algebra over k is its degree.

Proposition 1.5. *Let B be a semisimple k -algebra. For any faithful k -module M , $\dim_k M \geq [B, k]_{\text{red}}$.*

Sketch of proof. Let $B = \prod B_i$ where $B_i \cong M_{n_i}(D_i)$ for some central division algebra D_i over k_i . Then $S_i = D_i^{n_i}$ is a simple B_i -module. Every B -module M is isomorphic to a sum $\bigoplus m_i S_i$ and M is faithful if and only if $m_i \geq 1$ for any i . Then

$$\dim_k(M) \geq \sum m_i \cdot n_i [D_i : k] \geq \sum n_i [D_i : k_i] [k_i : k]$$

while $[B : k]_{\text{red}} = \sum n_i [D_i : k_i]^{1/2} [k_i : k]$ by definition. $\square$

Corollary 1.9. *For any abelian variety A ,*

$$2\dim A \geq [\text{End}^0(A) : \mathbb{Q}]_{\text{red}} \quad (1.1)$$

When equality holds, $\text{End}^0(A)$ is a product of matrix algebras over fields. In particular, if A is a simple abelian variety over a field $k \subseteq \mathbb{C}$, then $2\dim A \geq [\text{End}^0(A) : \mathbb{Q}]$

Proof. As $\text{End}^0(A)$ is a semisimple $\mathbb{Q}$ -algebra acting faithfully on the rational homology group $H_1(A, \mathbb{Q})$ which is a $\mathbb{Q}$ -vector space with dimension $2\dim A$, this follows from proposition 1.5 immediately. $\square$

Definition 1.12. *An abelian variety A is said to have **complex multiplication** (or be of CM-type, or be a CM abelian variety) if $2\dim A = [\text{End}^0(A) : \mathbb{Q}]_{\text{red}}$. Equivalently, $\text{End}^0(A)$ contains an étale $\mathbb{Q}$ -subalgebra of degree $2\dim A$.*

Remark 1.4. A corollary of Tate's isogeny theorem shows that all abelian varieties over finite fields are of CM type. See [EvdM, theorem 16.27].

Note that by corollary 1.9, in this case $\text{End}^0(A) = \prod M_{n_i}(k_i)$ and its center $Z = \prod k_i$ is smaller than the required étale $\mathbb{Q}$ -subalgebra of degree $2\dim A$ in general.

Proposition 1.6. (a) *A simple abelian variety A has complex multiplication if and only if $\text{End}^0(A)$ is a CM-field of degree $2\dim A$ over $\mathbb{Q}$.*

(b) *An isotypic abelian variety A has complex multiplication if and only if $\text{End}^0(A)$ contains a CM field of degree $2\dim A$ over $\mathbb{Q}$.*

(c) *An abelian variety A has complex multiplication if and only if $\text{End}^0(A)$ contains a CM $\mathbb{Q}$ -algebra of degree $2\dim A$ over $\mathbb{Q}$*

Proof. (a) If A has CM, then $\text{End}^0(A) = \prod M_{n_i}(k_i)$ has to be one of these k_i 's because $\text{End}^0(A)$ is a division algebra. It is a CM field because of Rosati involution and proposition 1.4

(b) Suppose $A \sim A_0^m$ with A_0 simple. Note that in this case $\text{End}^0(A_0)$ can still only be of the form in (a) and hence is a CM field and we denote $\text{End}^0(A_0) = E_0$. Let F be a totally real field of degree m over $\mathbb{Q}$ that is linearly disjoint from E_0 . Then $E = E_0 \cdot F$ is a CM-field of degree $2\dim A$, and the choice of an E_0 -basis for it defines an embedding of it into $M_m(E_0) \simeq \text{End}^0(A_0^m) = \text{End}^0(A)$

(c) follows from (b) and the decomposition theorem. $\square$

Definition 1.13. *Let E be a CM algebra. A **CM type** on E is a subset $\Phi \subseteq \text{Hom}(E, \mathbb{C})$ such that $\text{Hom}(E, \mathbb{C}) = \Phi \sqcup c\Phi$, where c is complex conjugation.*

So taking a CM type is the same as choosing a $\phi \in \text{Hom}(E, \mathbb{C})$ from each conjugate pair.

Definition 1.14. *Let E be a CM field. An abelian variety with CM by E is a tuple (A, i) , where A is an abelian variety of dimension $\frac{1}{2}[E : \mathbb{Q}]$ and $i : E \hookrightarrow \text{End}^0(A)$ is an embedding of E as a subfield of $\text{End}^0(A)$.*

For the dimension reason, A has CM by a CM field E is isotypic by proposition 1.6 (b)

Remark 1.5. The tuple (A, i) determines a CM type $\Phi \subseteq \text{Hom}(E, \mathbb{C})$ on E associated to (A, i) . Indeed, given $A/\mathbb{C}$ and $i : E \rightarrow \text{End}^0(A)$ for a CM field E , E acts on $\text{Tgt}_0(A)$, and we may decompose this representation as $\bigoplus_{\phi \in \Phi} \mathbb{C}_\phi$ with the canonical structure of E -module, where Φ is the CM type of (A, i) .

1.6 Frobenius Endomorphism

Definition 1.15. Let $V/\mathbb{F}_q$ be a variety. The Frobenius map $\pi_V : V \rightarrow V$ is defined to be the identity map on the underlying topological space of V and is the map $f \rightarrow f^q$ on $\mathcal{O}_V$.

Let A be an abelian variety over $\mathbb{F}_q$. Then π_A maps identity to identity and the base field is of characteristic p implies $\pi_A \in \text{End}(A)$. We write $f_A = P_{\pi_A}$ for the characteristic polynomial of π_A acting on $V_\ell A$.

Proposition 1.7. (W  il) For any $\alpha \in \text{End}(A)$ (and particularly π_A), P_α is a monic polynomial of degree $2g$, $g = \dim A$, with coefficients in $\mathbb{Z}$. In particular, π_A is an algebraic integer.

Proof. See [Mil08, theorem 10.9 and proposition 10.20] □

Lemma 1.3. $(\pi_A)^\dagger \circ \pi_A = [q]_A$

There are many different approaches to this, and we choose the fancy one by relative Frobenius $F_{A/\mathbb{F}_q} : A \rightarrow A^{(p)}$ and Verschiebung $V_{A/\mathbb{F}_q} : A^{(p)} \rightarrow A$ which will be introduced in section 5.3. By the definition of Verschiebung, we have $V_{A/\mathbb{F}_q}^\vee = F_{A^\vee/\mathbb{F}_q}$. Also $A^{(q)} = A$ since the base field is $\mathbb{F}_q$.

Now $(\pi_A)^\dagger \circ \pi_A = \lambda^{-1} \circ \pi_A^\vee \circ \lambda \circ \pi_A = \lambda^{-1} \circ \pi_A^\vee \circ \pi_{A^\vee} \circ \lambda$, so it suffices to show that $\pi_A^\vee \circ \pi_{A^\vee} = [q]_{A^\vee}$. By definition we notice that $\pi_A = F_{A^{(p^n-1)}} \circ \dots \circ F_A$ where $q = p^n$. Then

$$\pi_A^\vee \circ \pi_{A^\vee} = (F_{A^{(p^n-1)}} \circ \dots \circ F_A)^\vee \circ F_{A^\vee} = V_{A^\vee} \circ \dots \circ V_{A^\vee} \circ F_{A^\vee} = [q]_{A^\vee}$$

by the fact that $V \circ F = [p]$ in remark 5.5.

Theorem 1.6. Every complex root α of f_A has absolute value $|\alpha| = q^{\frac{1}{2}}$

Proof. We first reduce it to the case when A is simple. Choose an isogeny

$$h : A \rightarrow A' = \prod_{i=1}^n A_i$$

where the factors A_i are all simple. This induces an isomorphism

$$V_\ell(h) : V_\ell(A) \cong \bigoplus V_\ell(A_i)$$

Since h is an isomorphism in $\text{Hom}^0(A, A')$, we see that $V_\ell(\pi_A)$ is the same as $V_\ell(\pi_{A'})$ after base change, which acts as $V_\ell(\pi_{A_i})$ on each component $V_\ell(A_i)$. So $f_A = \prod_{i=1}^n f_{A_i}$, and it suffices to prove the theorem for simple abelian varieties.

We will see in theorem 2.4 that $L = \mathbb{Q}(\pi_A)$ is the center of $\text{End}^0(A)$ when A is simple. Hence $\dagger$ on $\text{End}^0(A)$ induces an involution on L , and moreover, $\dagger$ induces an involution $\dagger^\mathbb{R}$ on $L \otimes_\mathbb{Q} \mathbb{R}$. By proposition 1.4, this algebra is a finite product of copies of $\mathbb{R}$ and of $\mathbb{C}$. Moreover, $\dagger$ is positive definite implies that $\dagger^\mathbb{R}$ is a positive definite $\mathbb{R}$ -linear involution on this product. So $\dagger^\mathbb{R}$ is the identity on every real factor and is the complex conjugation on every complex factor as how $\dagger$ acts on L . Then we have

$$\forall x \in L, \forall \phi : L \hookrightarrow \mathbb{C}, \phi(x^\dagger) = \phi(x)^{\dagger\mathbb{R}} = \overline{\phi(x)}$$

Hence

$$q = \phi(q) = \phi((\pi_A)^\dagger \circ \pi_A) = \overline{\phi(\pi_A)} \cdot \phi(\pi_A) \Rightarrow |\phi(\pi_A)| = q^{\frac{1}{2}}$$

□

Definition 1.16. A Weil q -number π is an algebraic integer such that, for every embedding $\phi : \mathbb{Q}[\pi] \hookrightarrow \mathbb{C}$, $|\phi(\pi)| = q^{\frac{1}{2}}$

Remark 1.6. Theorem 1.6 shows that $\pi_A \in \text{End}(A)$ is a Wéil q -number.

The Frobenius π_A is important for an abelian variety A over a finite field. Not only π_A generates the center of A 's endomorphism algebras, its conjugate class also determines the isogeny class of A , as to be seen in Honda-Tate.

2 Tate's Isogeny Theorem

2.1 Statement and Proof

In this section we aim to prove Tate's theorem which characterizes the homomorphism rings of abelian varieties. Without further notification, we always assume $k = \mathbb{F}_q$ to be a finite field across this section.

Theorem 2.1. (*Tate's Isogeny Theorem*) Let $\ell \neq p$, and let $A, B/\mathbb{F}_q$ be abelian varieties. Then

$$\text{Hom}(A, B) \otimes_{\mathbb{Z}} \mathbb{Z}_\ell \cong \text{Hom}_{\Gamma_k}(T_\ell(A), T_\ell(B)).$$

The power of Tate's theorem is that when considering the homomorphisms between two abelian varieties we can instead look at the G -module linear maps between their Tate modules which are of course linear objects and hence easier to deal with to some extent. This is a central motivation for us to study the Galois representation $T_\ell(A)$.

For the proof, we start with some observations by extending scalars.

Lemma 2.1. *The map*

$$T_\ell : \text{Hom}(A, B) \otimes_{\mathbb{Z}} \mathbb{Z}_\ell \rightarrow \text{Hom}_{\Gamma_k}(T_\ell(A), T_\ell(B))$$

is injective with torsion-free cokernel.

Proof. We may assume $\text{Hom}(A, B)$ is finitely generated. Take $\{\phi_i\}$ a $\mathbb{Z}$ -basis for $\text{Hom}(A, B)$, and suppose $\sum \phi_i \otimes \lambda_i$ is mapped to zero. Let $b_i \in \mathbb{Z}$ such that $b_i \equiv \lambda_i \pmod{\ell^n}$ for a given n . Then $\sum b_i \phi_i : A \rightarrow B$ vanishes on $A[\ell^n]$ and hence factors through $[\ell^n]$, i.e. $\exists \varphi : A \rightarrow B$ such that $\sum b_i \phi_i = \ell^n \varphi$. Hence ℓ^n divides b_i and $\lambda_i \equiv 0 \pmod{\ell^n}$. Since n is arbitrary, λ_i 's are all zero, and so T_ℓ is injective.

Let $\psi \in \text{Hom}_{\Gamma_k}(T_\ell(A), T_\ell(B))$. Suppose $\exists \lambda \in \mathbb{Z}_\ell$ such that $\lambda \psi = T_\ell(\sum \phi_i \otimes \lambda_i)$ as in above settings. Take n to be the maximal such that $\lambda \equiv 0 \pmod{\ell^n}$. Then $\lambda = \ell^n \lambda'$ with λ' a unit in $\mathbb{Z}_\ell$. Moreover, $\lambda \psi$ (and hence $\sum b_i \phi_i$ where $b_i \equiv \lambda_i \pmod{\ell^n}$) vanishes on $A[\ell^n]$, which implies that $\lambda_i \equiv 0 \pmod{\ell^n}$ as before. Let $\lambda_i = \ell^n \lambda'_i$. Then $\psi = T_\ell(\sum \phi_i \otimes \lambda'^{-1} \lambda'_i)$ and so the cokernel of T_ℓ is torsion-free. □

Corollary 2.1. *With the above notations, T_ℓ is an isomorphism if and only if the map*

$$V_\ell : \text{Hom}^0(A, B) \otimes_{\mathbb{Q}} \mathbb{Q}_\ell \rightarrow \text{Hom}_{\Gamma_k}(V_\ell(A), V_\ell(B)) \tag{2.1}$$

is an isomorphism.

Proof. As $\mathbb{Q}_\ell$ is flat over $\mathbb{Z}_\ell$, V_ℓ is also injective with $\text{coker}(V_\ell) = \text{coker}(T_\ell) \otimes \mathbb{Q}_\ell$. Note that $\text{coker}(T_\ell) = 0$ if and only if $\text{coker}(T_\ell) \otimes \mathbb{Q}_\ell = 0$ since $\text{coker}(T_\ell)$ is torsion-free by lemma 2.1, and the statement follows. $\square$

Lemma 2.2. *Assume that for every abelian variety C over k , the map*

$$\text{End}^0(C) \otimes_{\mathbb{Q}} \mathbb{Q}_\ell \rightarrow \text{Aut}_{\Gamma_k}(V_\ell C) \quad (2.2)$$

is an isomorphism, then for any two abelian varieties A and B over k , the map in (2.1) is an isomorphism.

Indeed, take $C = A \times B$, and we have the decomposition of vector spaces

$$\text{End}^0(C) = \text{End}^0(A) \oplus \text{Hom}^0(A, B) \oplus \text{Hom}^0(B, A) \oplus \text{End}^0(B)$$

and likewise for $\text{End}_{\Gamma_k}(V_\ell C)$. Note that the map in (2.2) respects these decompositions. Hence if it is an isomorphism then so is the map in (2.1) which is now a component. Therefore we have reduced it to only consider endomorphism rings.

Here is a key observation by Tate:

Theorem 2.2. *There exists only a finite number of isomorphism classes of abelian varieties B defined over k such that there exists a k -isogeny $B \sim A$ of ℓ -power degree.*

Faltings further showed that for A an abelian variety over a number field K , there are only finitely many isomorphism classes of abelian varieties over K isogenous to A , which is included in his proof of Mordell's Conjecture. See the discussion in [Mil08, chapter V]. However, even the proof of the original version over finite fields is beyond the scope of this essay, which can be found in [FM82]

One can prove the following lemma with this observation:

Lemma 2.3. *Let A an abelian variety over a field k , and let ℓ be a prime number such that $\ell \neq \text{char}(k)$. Then for every $\mathbb{Q}_\ell$ -subspace $W \subset V_\ell A$ that is stable under the action of $G = \Gamma_k$ there exists an element $u \in \mathbb{Q}_\ell \otimes \text{End}^0(A)$ such that $W = u \cdot V_\ell A$.*

Sketch of proof. For $n \in \mathbb{Z}_{\geq 1}$, let $U_n = (W \cap T_\ell(A)) + \ell^n T_\ell(A)$ and G_n be its image in $T_\ell(A)/\ell^n T_\ell(A) \cong A[\ell^n](\bar{k})$. Then G_n can be considered as the $\bar{k}$ -points of a subgroup scheme (which we will still denote by G_n) of $A[\ell^n]$ by the fact that G_n is Γ_k -stable and étale k -group schemes are in 1-to-1 correspondence with Γ_k -groups (see theorem 5.2). Let $A_n = A/G_n$ and $f_n : A \rightarrow A_n$ be the quotient map. As $A_n \subset A[\ell^n]$, $[\ell^n]$ factors through A_n , i.e. $\exists$ unique isogeny $g_n : A_n \rightarrow A$ such that $g_n \circ f_n = [\ell^n]_A$. The kernel of g_n is a quotient of $A[\ell^n]$, and hence the degree of g_n is ℓ -power. Moreover, one can show that $U_n \cong T_\ell(A_n)$ as Galois representations by analyzing the induced map by g_n between their Tate modules.

Now by the finiteness theorem, we can find an increasing sequence $n = n_1 < n_2 < \dots$ such that there are isomorphisms $\alpha_i : A_{n_i} \xrightarrow{\sim} A_{n_{i+1}}$, and we define $u_i = g_{n_i} \circ \alpha_i \circ f_n$ which are endomorphisms of A . The induced map $T_\ell(u_i)$ of $T_\ell(A)$ is the composition

$$T_\ell(A) \xrightarrow{\cdot \ell^n} U_n \xrightarrow{T_\ell(\alpha_i)} U_{n_i} \hookrightarrow T_\ell(A)$$

Observe that $\text{End}(A) \otimes_{\mathbb{Z}} \mathbb{Z}_\ell$ is a free $\mathbb{Z}_\ell$ -module of finite rank and hence compact for the ℓ -adic topology. Hence we may replace the sequence n_i by a subsequence to assume that they converge to some $u \in \text{End}(A) \otimes_{\mathbb{Z}} \mathbb{Z}_\ell$. Then $T_\ell(u)$ maps $T_\ell(A)$ to $\cap_{i \geq 1} U_{n_i} = W \cap T_\ell(A)$. Take $x \in W \cap T_\ell(A)$. Then $\ell^n x$ lies

in the image of $T_\ell(u_i)$ for all i , which implies that it is arbitrarily close to the image of $T_\ell(u)$. But this image is closed and so $\ell^n x \in \text{Im}(T_\ell(u))$ and $\ell^n(W \cap T_\ell(A)) \subseteq \text{Im}(T_\ell(u))$. After extending to $\mathbb{Q}_\ell$, we see that the image of $V_\ell(u)$ is precisely W . $\square$

With this lemma the proof of Tate's theorem is easy. Recall that the **graph** Γ of $\alpha \in \text{End}_R(M)$ is $\Gamma(\alpha) = \{(x, \alpha(x)) : x \in M\} \subseteq M \oplus M$,

Proof of main theorem. Take $\alpha \in \text{Aut}_{\Gamma_k}(V_\ell A)$. α is Γ_k -equivalent implies that the graph Γ of α is a Γ_k -stable subspace of $V_\ell A \times V_\ell A = V_\ell(A \times A)$. By lemma 2.3, there is an endomorphism induced by $u \in \text{End}(A \times A) \otimes \mathbb{Q}_\ell$ whose image is Γ , and it can be identified as a 2×2 matrix. Pick an element c of the centralizer Z of $\text{End}(A) \otimes \mathbb{Q}_\ell$ in $\text{Aut}_{\Gamma_k}(V_\ell A)$. Then the diagonal matrix $\text{diag}(c, c)$ commutes with u and therefore

$$\text{diag}(c, c)(\Gamma) = \text{diag}(c, c) \circ u(V_\ell A \times V_\ell A) = u \circ \text{diag}(c, c)(V_\ell A \times V_\ell A) \subseteq u(V_\ell A \times V_\ell A) = \Gamma$$

So if $(x, \alpha(x)) \in \Gamma$ then $(c(x), c(\alpha(x))) \in \Gamma$. Since Γ is the graph of α , also $(c(x), \alpha(c(x))) \in \Gamma$. This implies $c \circ \alpha = \alpha \circ c$ and hence α is in the centralizer of Z , which is $\text{End}^0(A) \otimes \mathbb{Q}_\ell$ by the double centralizer theorem. This shows the surjectivity of (2.2). $\square$

Remark 2.1. Tate's original proof is more complicated by induction on the dimension of the maximal Γ_k -stable subspace. This proof is based on [EvdM, theorem 16.18], where they also showed the semi-simplicity of the representation $\rho_\ell : \Gamma_k \rightarrow \text{Aut}(V_\ell A)$ using the finiteness theorem.

Remark 2.2. One would ask if Tate's theorem still holds for abelian varieties over number fields. More on this can be found in [MW].

2.2 Application

The most important results from Tate's theorem are the following

Theorem 2.3. *Let A and B be abelian varieties over a finite field k . TFAE:*

- (1) A and B are isogenous
- (2) $V_\ell A$ and $V_\ell B$ are G -isomorphic for some $\ell \neq \text{char}(k)$
- (3) $f_A = f_B$

Proof. (1) $\Rightarrow$ (2) $\Rightarrow$ (3) is clear, and (3) $\Rightarrow$ (2) because a semi-simple representation is determined by its characteristic polynomial. For (2) $\Rightarrow$ (1), suppose $u : V_\ell A \rightarrow V_\ell B$ is a G -isomorphism, by (2.1) we can find $\phi \in \text{Hom}^0(A, B) \otimes_{\mathbb{Q}} \mathbb{Q}_\ell$ such that $V_\ell(\phi) = u$. We can approximate it by elements in $\text{Hom}^0(A, B)$ which, if close enough, will also induce an isomorphism between $V_\ell A$ and $V_\ell B$. Multiplying by an integer we obtain a ϕ' in $\text{Hom}(A, B)$ inducing an isomorphism $V_\ell A \rightarrow V_\ell B$. If $\ker \phi'$ contains any positive-dimensional abelian variety C , then $V_\ell(\phi')$ would annihilate $V_\ell C$. Hence ϕ' has finite kernel. Similarly, one can show that it is surjective and hence an isogeny. $\square$

Remark 2.3. More generally, B is k -isogenous to an abelian subvariety of A if and only if $V_\ell(B)$ is G -isomorphic to a G -subspace of $V_\ell(A)$ for some $\ell \neq \text{char}(k)$ if and only if f_B divides f_A .

The next results gives a precise description of the structure of endomorphism algebras of isotypic abelian varieties.

Theorem 2.4. *Let A be abelian variety of dimension g over $\mathbb{F}_q$.*

- (a) $F = \mathbb{Q}[\pi_A]$ is the center of $D = \text{End}^0(A)$. In particular, A is isotypic if and only if $\mathbb{Q}[\pi_A] = \mathbb{Q}(\pi_A)$ is a field
- (b) Suppose A is isotypic, let m_A be the minimal polynomial of π_A over $\mathbb{Q}$. Let $d = [D : F]^{1/2}$ and $e = [F : \mathbb{Q}]$. Then $2g = de$ and $f_A = (m_A)^d$
- (c) For a place v of F , $\text{inv}_v(D) = \frac{\text{ord}_v(\pi_A)}{\text{ord}_v(q)} [F_v : \mathbb{Q}_p]$ if $v \mid p$, $\text{inv}_v(D) = \frac{1}{2}$ if v is real, and 0 otherwise.

Proof. See [EvdM, theorem 16.26] □

3 Good Reduction of Abelian Varieties with CM

3.1 Good Reduction and Néron Model

Definition 3.1. *Let A be an abelian variety over a field K . Let v be a discrete valuation of K with R the valuation ring and k the residue field. We say that A has **good reduction** at v if there exists an abelian scheme $\mathcal{A}$, i.e. a smooth proper group scheme over $\text{Spec}(R)$ with generic fiber $\mathcal{A} \times_R K \simeq A$. From now on we denote the generic fibre by $\mathcal{A}_K$ and the special fibre $\mathcal{A} \times_R k$ by $\mathcal{A}_k$*

Given an abelian variety A over a number field K with ring of integers $\mathcal{O}_K$, a standard argument shows that A extends to an abelian scheme over a non-empty open subset S of $\text{Spec}(\mathcal{O}_K)$, i.e. A has good reduction at all points of S . In general, one cannot expect that A also has good reduction at the finitely many points in $\text{Spec}(R) - S$. However, after removing the condition of properness, Néron discovered that a somewhat still nice model always exists.

Definition 3.2. *A **Néron model** $\mathcal{A}$ of A over R is a smooth separated R -scheme of finite type such that:*

1. $\mathcal{A}_K \cong A$;
2. (*Néron mapping property*) For every smooth R -scheme X and a K -morphism $\phi_K : X_K \rightarrow A$, there exists a unique map $\phi : X \rightarrow \mathcal{A}$ extending ϕ_K .

Theorem 3.1. *Any abelian variety A/K admits a Néron model $\mathcal{A}$.*

In [BLR90] they gave an exposition of Néron's construction in the language of schemes. The most important characteristic of Néron model is its universal property. To show that the models induced by good reduction are Néron models, we need the following two lemmas from algebraic geometry:

Lemma 3.1. *Let $S = \text{Spec} R$, with R a DVR, and let G be a smooth group scheme over S . For any smooth scheme X over S and rational map $f : X \dashrightarrow G$, the set of points where f is not defined has pure codimension one in X .*

Proof. See [Mil20, theorem 6.5] □

Lemma 3.2. *Let $Y \rightarrow S$ be a proper map, and let $X \rightarrow S$ be a smooth map; every S -morphism $X \setminus Z \rightarrow Y$ with Z of pure codimension 1 in X extends uniquely to an S -morphism $X \rightarrow Y$.*

Proof. Let ξ be the generic point of an irreducible component of Z . Since X is smooth and Z of pure codimension 1, $\mathcal{O}_{X,\xi}$ is regular of dimension 1 and hence is a DVR. Let $K = \text{Frac}(\mathcal{O}_{X,\xi})$ and ξ' be the generic point of the irreducible component of X containing ξ . Then $\xi' \in X \setminus Z$ as Z has pure codimension 1. Note ξ' is a K -point, and it gives a map $\text{Spec}(K) \hookrightarrow X \setminus Z \rightarrow Y$. By properness of $Y \rightarrow S$, $\text{Spec}(K) \rightarrow Y$ admits a unique extension $\text{Spec}(\mathcal{O}_{X,\xi}) \rightarrow Y$. So ξ is defined. (Locally, properness gives that $\varphi : \text{Spec}(K) \hookrightarrow \text{Spec}(B)$ factors through each $\text{Spec}(A_p)$ for p of height 1. So $\varphi^\# : B \rightarrow K$

factors through A_p for each height-1 p . Using the fact that $\bigcap_{\text{height}=1 p} A_p = A$, we see that $\varphi^\#$ lands in A .) $\square$

Proposition 3.1. *Let $\mathcal{A}$ be an abelian scheme over a DVR R with generic fibre A . For any smooth R -scheme X , every morphism $X_K \rightarrow A$ extends uniquely to a morphism $X \rightarrow \mathcal{A}$.*

Proof. We assume X is reduced. Note that X is smooth and hence flat over R . Then [Har77, Chapter III proposition 9.7] says that the generic points of irreducible components of X maps to the generic point of $\text{Spec}(R)$. So X_K is dense open in X . Then $X_K \rightarrow A \rightarrow \mathcal{A}$ gives a class of rational map $X \dashrightarrow \mathcal{A}$. Now the statement follows from these two lemmas directly. $\square$

Corollary 3.1. *Let A and B be abelian varieties over K , and let $\mathcal{A}$ and $\mathcal{B}$ be abelian schemes over R with generic fibres A and B respectively. The restriction map*

$$\text{Hom}_R(\mathcal{A}, \mathcal{B}) \rightarrow \text{Hom}_K(A, B)$$

is a bijection. In particular, B (if it exists) is unique up to a unique isomorphism.

Proof. Proposition 3.1 gives an inverse to the restriction map $\text{Mor}_R(\mathcal{A}, \mathcal{B}) \rightarrow \text{Mor}_K(A, B)$, and it is easy to see that homomorphisms correspond to homomorphisms $\square$

Corollary 3.1 shows that the models given by good reduction have the Néron mapping property. It is equivalent to say that the "taking the generic fibre" functor from the category of abelian schemes over R to categories of abelian varieties over K is fully faithful.

Since we have been playing with homomorphisms, the functor of points perspective could be helpful. For instance, the m -torsion K -points of A are essentially $h_A(\mathbb{Z}/m\mathbb{Z}) = \text{Hom}_K(\mathbb{Z}/m\mathbb{Z}, A)$. This gives

Corollary 3.2. $\mathcal{A}(R) = A(K)$. In particular, $\mathcal{A}(R)[m] = A(K)[m]$.

Proof. This is clear by the Néron mapping property. $\square$

Proposition 3.2. *Let A be an abelian variety over K with good reduction at v . If R the valuation ring w.r.t. v is henselian (i.e. where Hensel's lemma holds), then for every m prime to the characteristic of k , there is an isomorphism $A(K)[m] \rightarrow \mathcal{A}(k)[m]$.*

Proof. By corollary 3.2, the reduction map $R \rightarrow k$ defines a homomorphism $r : A(K)[m] = \mathcal{A}(R)[m] \rightarrow \mathcal{A}(k)[m]$. Since R is henselian and $\mathcal{A}$ is smooth, r is surjective by [BLR90, proposition 2.3.5]. Moreover, since m is prime to $\text{char}(k)$, $[m]$ is an étale endomorphism of $\mathcal{A}$. Combining with the assumption that R is Henselian, this shows that the kernel of r is uniquely divisible by m . Hence r gives an isomorphism $A(K)[m] \xrightarrow{\sim} \mathcal{A}(k)[m]$ $\square$

To apply proposition 3.2, we extend the valuation v on K to an algebraic closure $\overline{K}$ of K , and let $I \subset D \subset \Gamma_K$ be the inertia and decomposition groups w.r.t v . Then $D/I \cong \text{Gal}(k^{\text{sep}}/k)$, and the ring of integers of $K^{ur} = \overline{K}^I$ is henselian, whose residue field is exactly $k^{\text{sep}} = \overline{k}$ since we always assume k to be perfect.

Corollary 3.3. *If A over K has good reduction, $A(K^{ur})[m] = A(\overline{K})[m]$ for all m prime to the characteristic of k*

Proof. By proposition 3.2, $\#A(K^{ur})[m] = \#\mathcal{A}_k(\overline{k})[m] = m^{2\dim A} = \#A(\overline{K})[m]$ since A/K has good reduction and hence $\mathcal{A}_k$ is an abelian variety over k . But it is clear that $A(K^{ur})[m] \subseteq A(\overline{K})[m]$. $\square$

Remark 3.1. Corollary 3.4 shows that the inertia group acts trivially on $A(\overline{K})[m]$, and hence trivially on $T_\ell A$ for $\ell \neq p$. This implies that all ℓ -torsion $\overline{K}$ -points are defined over an unramified extension of K for ℓ prime to p .

3.2 Potentially Good Reduction and the Néron-Ogg-Shafarevich Criterion

It turns out that the converse of corollary 3.3 also holds, and it gives the following criterion for good reduction:

Theorem 3.2. (*Néron-Ogg-Shafarevich Criterion*) *Let R be a DVR with fraction field K and residue field k . Let $\ell \neq \text{char}(k)$ be a prime, and let A/K be an abelian variety. Then A has good reduction if and only if the inertia group $I \subseteq \Gamma_K$ acts trivially on $T_\ell(A)$ (In this case we say $T_\ell A$ is unramified.)*

Sketch of proof. Remark 3.1 has shown that good reduction implies that I acts trivially on $T_\ell(A)$. Conversely, let $\mathcal{A}/R$ be the Néron model of A/K . It now suffices to show that $\mathcal{A}$ is proper. By [ST68, lemma 3], this is again reduced to show that the special fibre $\mathcal{A}_k$ is an abelian variety.

We know that $\mathcal{A}_k$ is a smooth algebraic group over k . Let $\mathcal{A}_k^o$ be the connected component of the identity. By the summary part of [Mil17], the quotient $\mathcal{A}_k/\mathcal{A}_k^o$ is a finite group and the connected algebraic group $\mathcal{A}_k^o$ admits a short exact sequence

$$1 \rightarrow U \rightarrow \mathcal{A}_k^o \rightarrow G \rightarrow 1$$

where U is unipotent and G is semi-abelian, i.e. G is an extension of a torus T by an abelian variety B :

$$1 \rightarrow T \rightarrow G \rightarrow B \rightarrow 1$$

Then $\dim A = \dim \mathcal{A}_k = \dim U + \dim T + \dim B$. We also have isomorphisms

$$A(K^{ur})[\ell^n] \cong \mathcal{A}(\mathcal{O}_{K^{ur}})[\ell^n] \cong \mathcal{A}_k(\bar{k})[\ell^n]$$

by proposition 3.2. We may also write

$$A(K^{ur})[\ell^n] = (A(\bar{K})[\ell^n])^I$$

So that the inertia group acts trivially on $T_\ell A$ implies that this is just $A(\bar{K})[\ell^n]$, and hence all these groups have order $\ell^{2\dim A}$.

We also have $\#|U(\bar{k})[\ell^n]| = \{1\}$, $\#|B(\bar{k})[\ell^n]| = \ell^{2\dim B}$, $\#|T(\bar{k})[\ell^n]| = \ell^{\dim T}$ using the fact that $T[\ell^n] = \mathbb{G}_m^{\dim T}[\ell^n] = (\mu_{\ell^n})^{\dim T}$ and unipotent groups are torsion-free.

Then $\ell^{2\dim A} = \#|\mathcal{A}_k(\bar{k})[\ell^n]| = \#|\mathcal{A}_k/\mathcal{A}_k^o(\bar{k})[\ell^n]| \cdot \#|B(\bar{k})[\ell^n]| \cdot \#|T(\bar{k})[\ell^n]| = (\text{some bounded number}) \times \ell^{n(2\dim B + \dim T)}$ for all n . Let $n \rightarrow \infty$ and we get

$$2\dim A = 2\dim B + \dim T$$

Comparing these two dimension formulas, we get U and T are both trivial and $\mathcal{A}_k^o = G = B$ is an abelian variety. So $\mathcal{A}_k$ is an abelian variety. $\square$

Definition 3.3. *Let A be an abelian variety over a number field K , and let p be a prime of K with the induced valuation v . We say that A has **potentially good reduction** at p if there exists a finite extension L/K and a valuation $v' \mid v$ such that A_L has good reduction at v' .*

A over a number field K always has good reduction at all but finitely many primes p . Therefore, if A has potentially good reduction at all primes, then there is a finite extension L/K such that A has good reduction everywhere by taking the compositum of all of the individual extensions since good reduction is stable under field extension. Indeed, we now show that all abelian varieties with CM have this property

Corollary 3.4. *Assume that K is a complete local field. Suppose that for some $\ell \neq p = \text{char}(k)$, the image of $\rho_\ell : \Gamma_K \rightarrow \text{Aut}(T_\ell(A))$ is commutative. Then A has potential good reduction*

Proof. Because the image of ρ_ℓ is commutative, ρ_ℓ factors through $\overline{\rho}_\ell : \Gamma_K^{ab} \rightarrow \text{Aut}(T_\ell(A))$, and we have a diagram

$$\begin{array}{ccc} I^{ab} & \xrightarrow{\overline{\rho}_\ell} & \text{Aut}(T_\ell(A)) \xleftarrow{i} 1 + \ell \cdot \text{End}(T_\ell(A)) \\ \theta \uparrow & & \uparrow \text{---} \\ \mathcal{O}_K^\times & \xleftarrow{i} & 1 + p\mathcal{O}_K \end{array}$$

where θ is the local Artin map, both inclusions are of finite index and all arrows are continuous homomorphisms.

Note that the group $1 + p\mathcal{O}_K$ is a pro- p group. Since $\ell \neq p$, the image of $1 + p\mathcal{O}_K$ in $\text{Aut}(T_\ell(A))$ must have trivial intersection with the pro- ℓ group $1 + \ell \cdot \text{End}(T_\ell(A))$. Hence this image is finite in $\text{Aut}(T_\ell(A))$ since it descends injectively to the finite quotient $\text{Aut}(T_\ell(A))/(1 + \ell \cdot \text{End}(T_\ell(A)))$. Because $1 + p\mathcal{O}_K$ is of finite index in $\mathcal{O}_K^\times$, the image $\rho_\ell(I) = \overline{\rho}_\ell \circ \theta(\mathcal{O}_K^\times)$ is also finite. By [ST68, lemma 2 (i)], there exists a finite extension L such that the inertia group of that field acts trivially on $T_\ell(A_L)$. By the Néron-Ogg-Shafarevich criterion, A has potentially good reduction to L . $\square$

Proposition 3.3. *If A is an abelian variety with CM over a number field K , then A has potentially good reduction everywhere.*

Proof. Assume that A has complex multiplication by $E \hookrightarrow \text{End}^0(A)$ over K . Let $R = E \cap \text{End}(A)$ which is an order of E and $\rho_\ell : \Gamma_K \rightarrow \text{Aut}(T_\ell(A))$ be an attached ℓ -adic representation for some ℓ . By corollary 3.4, it suffices to show that the image of ρ_ℓ is commutative and then the image of ρ_ℓ restricted to the decomposition group of each prime is also commutative. Because the elements in R are defined over K , they commute with the action of Γ_K . Hence the image of ρ_ℓ lies in the centralizer of R in $\text{Aut}(T_\ell A)$. But [Mil20, theorem 7.4] tells us that the centralizer of R in $\text{Aut}(T_\ell(A))$ is $R \otimes_{\mathbb{Z}} \mathbb{Z}_\ell$, which is commutative. $\square$

Notice that the order $R = E \cap \text{End}(A)$ is a subring of $\mathcal{O}_E$. An important fact about R is that it does not change under base field extensions, i.e. $R = E \cap \text{End}(A_{k'})$ for any extension k'/k . See [Mil20, pp53].

We note that the CM type is also unaffected under base field extension k'/k for an abelian variety of CM by (E, Φ) , which is easily seen since $Tgt_0 A_{k'} = Tgt_0 A \otimes_k k' = \prod_{\phi \in \Phi} k_\phi \otimes_k k' = \prod_{\phi \in \Phi} k'_\phi$

3.3 The Shimura-Taniyama Formula

The Shimura-Taniyama formula characterizes how the Frobenius behaves in the endomorphism ring:

Theorem 3.3. *Let A be an abelian variety with CM by (E, Φ) over a number field K . Suppose that K contains the Galois closure of E and let $\mathfrak{P}$ be a prime ideal of $\mathcal{O}_K$ at which A has good reduction with the integral model $\mathcal{A}$ and $k = \mathcal{O}_K/\mathfrak{P} = \mathbb{F}_q$. Assume that $K_{\mathfrak{P}}/\mathbb{Q}_p$ is unramified in E and $\text{End}(A) \cap E = \mathcal{O}_E$ is the maximal order of E .*

- (a) *There exists an element $\pi \in \mathcal{O}_E$ inducing the Frobenius endomorphism on the $\mathcal{A}_k$.*
- (b) *The idea generated by π factors as follows,*

$$(\pi) = \prod_{\phi \in \Phi} \phi^{-1}(Nm_{K/\phi E} \mathfrak{P})$$

To prove (a), we need the following lemma on fibres:

Lemma 3.3. *If A/K has good reduction at $\mathfrak{P}$, then the natural map $\text{End}(A) \cong \text{End}(\mathcal{A}) \rightarrow \text{End}(\mathcal{A}_k)$ is injective*

This follows from [FM82, proposition 6.1], which says that for an S -morphism $f : X \rightarrow Y$ with S connected and X flat and proper over S , if a fibre X_s over some $s \in S$ is mapped to a (set-theoretically) single point by f , then f factors through a section $p : S \rightarrow Y$. Moreover, if X, Y are group schemes and f is homomorphism, then p has to be the unit section of Y , which implies that f is the identity map.

Proof of Theorem 3.3. (a) By the lemma, we have $\text{End}(A) \hookrightarrow \text{End}(\mathcal{A}_k)$, so we obtain

$$E \otimes \mathbb{Q}_\ell \subseteq \text{End}^0(A) \otimes \mathbb{Q}_\ell \hookrightarrow \text{End}^0(\mathcal{A}_k) \otimes \mathbb{Q}_\ell \simeq \text{Aut}_{\Gamma_k}(V_\ell \mathcal{A}_k)$$

by Tate's theorem, so $E \otimes \mathbb{Q}_\ell$ acts faithfully on $V_\ell \mathcal{A}_k$. Note that $V_\ell \mathcal{A}_k$ has $\mathbb{Q}_\ell$ -dimension $2\dim A$ while $E \otimes \mathbb{Q}_\ell$ is also a $\mathbb{Q}_\ell$ -algebra of $\mathbb{Q}_\ell$ -dimension $2\dim A$. So $V_\ell \mathcal{A}_k$ is a free rank-1 $E \otimes \mathbb{Q}_\ell$ -module. Hence the Frobenius $\pi_{\mathcal{A}_k}$ acts on $V_\ell \mathcal{A}_k$ as scalar multiplication by an element in $E \otimes \mathbb{Q}_\ell$. Now note that $(E \otimes \mathbb{Q}_\ell) \cap \text{End}(\mathcal{A}_k)$ is an order in E containing $\text{End}(A) \cap E = \mathcal{O}_E$. So they must agree and we may identify the Frobenius as an element $\pi \in \mathcal{O}_E$.

For (b) we only give a sketch of the proof. By lemma 1.3, $[q]_{\mathcal{A}_k}$ factors through $\pi \in \mathcal{O}_E$, which is identified with the Frobenius on $\mathcal{A}_k$. So π divides q , and we may write $(\pi) = \prod_{v|p} \mathfrak{p}_v^{m_v}$. We will see the norm of each factor is completely determined by the CM type Φ .

Let h be the class number of E . So $\mathfrak{p}_v^{hm_v} = (\gamma_v)$ for some $\gamma_v \in \mathcal{O}_E \subset \text{End}(\mathcal{A}_k)$. So γ_v can be seen as an endomorphism of $\mathcal{A}_k$. And we want to compute its degree.

Lemma 3.4. *For $\alpha \in \mathcal{O}_E$, the morphism $\alpha : \mathcal{A}_k \rightarrow \mathcal{A}_K$ is an isogeny with degree $Nm_{E/\mathbb{Q}}(\alpha)$*

If we consider α acting on $V_\ell(\mathcal{A}_k)$ for some ℓ , then $\deg(\alpha) = \det(\alpha|_{V_\ell(A)}) = Nm_{E/\mathbb{Q}}(\alpha)$. See [Mil08, lemma 12.9]. By the lemma we have $\deg(\gamma_v) = Nm_{E/\mathbb{Q}}(\mathfrak{p}_v^{hm_v})$. We also have a different way to evaluate the degree of an isogeny:

Lemma 3.5. *Let k be an algebraically closed field of characteristic p , and let $\alpha : A \rightarrow B$ be an isogeny of abelian varieties over k . Assume that $k(A)^q \subset \alpha^*(k(B))$ for some $q = p^m$, and let d be the dimension of the kernel of the induced map $\alpha : \text{Tgt}_0 A \rightarrow \text{Tgt}_0 B$. Then $\deg(\alpha) \leq q^d$*

See [Mil20, proposition 7.17]. Now we take $(e_\phi)_{\phi \in \Phi}$ a K -basis for $\text{Tgt}_0 A = \prod_{\phi \in \Phi} K_\phi$. Then $K_{\mathfrak{P}}/\mathbb{Q}_p$ is unramified implies that $(e_\phi)_{\phi \in \Phi}$ also can be chosen to be a $\mathcal{O}_{\mathfrak{P}}$ -basis for $\text{Tgt}_0 \mathcal{A}$, and hence $\text{Tgt}_0 \mathcal{A}_k = \text{Tgt}_0 \mathcal{A}/\mathfrak{P} \text{Tgt}_0 \mathcal{A}$ admits a k -basis $(\bar{e}_\phi)_{\phi \in \Phi}$ and $\ker(\gamma_v : \text{Tgt}_0 \mathcal{A}_k \rightarrow \text{Tgt}_0 \mathcal{A}_k)$ is spanned by those $\bar{e}_\phi$ with $\phi(\gamma_v) \in \mathfrak{P}$.

Since we assume that K contains the Galois closure of E , any embedding $K_{\mathfrak{P}} \hookrightarrow \overline{\mathbb{Q}_p} \hookrightarrow \mathbb{C}$ identifies $\text{Hom}(E_v, K_{\mathfrak{P}}) = \text{Hom}(E_v, \overline{\mathbb{Q}_p}) = \text{Hom}(E_v, \mathbb{C})$. We then define $H_v = \text{Hom}(E_v, \overline{\mathbb{Q}_p}) = \text{Hom}(E_v, K_{\mathfrak{P}}) = \{\tau \in \text{Hom}(E, K) | \tau^{-1}(\mathfrak{P}) = \mathfrak{p}_v\}$ and $\Phi_v = \Phi \cap H_v$.

We claim that $\bar{e}_\phi \in \ker(\gamma_v)$ if and only if $\phi \in \Phi_v$. If $\gamma_v \in \phi^{-1}(\mathfrak{P})$, then $\phi^{-1}(\mathfrak{P})$ is prime and γ_v is only divided by $\mathfrak{p}_v$ implies that $\phi^{-1}(\mathfrak{P}) = \mathfrak{p}_v$ and so $\phi \in \Phi_v$. The converse direction is clear. So $\dim_k(\ker(\gamma_v)) = \#\Phi_v$.

We now apply lemma 3.5 to γ_v . Since $\mathfrak{p}_v^{m_v}$ divides π , we see that π^h factors through γ_v by our construction. Since π^h is the q^h -th-power Frobenius, by definition $(\pi^h)^* k(\mathcal{A}_K) = k(\mathcal{A}_K)^{q^h}$ and hence $(\gamma_v)^* k(\mathcal{A}_K) \supset k(\mathcal{A}_K)^{q^h}$. By lemma 3.5, $Nm_{E/\mathbb{Q}}(\mathfrak{p}_v^{hm_v}) = \deg(\gamma_v) \leq q^{h\#\Phi_v}$. Take the h -th root and we get $Nm_{E/\mathbb{Q}}(\mathfrak{p}_v^{m_v}) \leq q^{\#\Phi_v}$ and hence

$$Nm_{E/\mathbb{Q}}(\pi) = Nm_{E/\mathbb{Q}} \prod_{v|p} \mathfrak{p}_v^{m_v} \leq \prod_{v|p} q^{\#\Phi_v} = q^{\#\Phi} = q^{\dim A}$$

But lemma 3.4 gives $Nm_{E/\mathbb{Q}}(\pi) = \deg(\pi : \mathcal{A}_k \rightarrow \mathcal{A}_k) = q^{\dim A}$. This is clear by noticing how π acts on the transcendence basis of $k(\mathcal{A}_k)$. So the above inequalities must all be equalities, and we get

$$Nm_{E/\mathbb{Q}} \mathfrak{p}_v^{m_v} = q^{|\Phi_v|} = (Nm_{K/\mathbb{Q}} \mathfrak{P})^{|\Phi_v|}$$

for each $v \mid p$. And this gives

$$\prod_{\phi \in \Phi_v} Nm_{K/\mathbb{Q}} \mathfrak{P} = Nm_{E/\mathbb{Q}} \left(\prod_{\phi \in \Phi_v} \phi^{-1}(Nm_{K/\phi E} \mathfrak{P}) \right)$$

Since $\prod_{\phi \in \Phi_v} \phi^{-1}(Nm_{K/\phi E} \mathfrak{P})$ is a power of $\mathfrak{p}_v$ by definition, this shows that

$$\mathfrak{p}_v^{m_v} = \prod_{\phi \in \Phi_v} \phi^{-1}(Nm_{K/\phi E} \mathfrak{P})$$

By taking the product over $v \mid p$, we obtain the required formula. $\square$

Here are two simple corollaries of Shimura-Taniyama formula that can be useful when the frobenius is involved:

Corollary 3.5. *With the above settings, for each prime $\mathfrak{p}_v$ of E dividing p ,*

$$ord_v(\pi) = \sum_{\phi \in \Phi_v} f(\mathfrak{P}/\phi \mathfrak{p}_v)$$

Proof. For each prime $\mathfrak{p}_v$ of E dividing p and $\phi \in \Phi$, if $\mathfrak{p}_v = \phi^{-1}(\mathfrak{P})$, then

$$ord_v(\phi^{-1}(Nm_{K/\phi E} \mathfrak{P})) = ord_{\phi \mathfrak{p}_v}(Nm_{K/\phi E} \mathfrak{P}) = f(\mathfrak{P}/\phi \mathfrak{p}_v)$$

and zero other wise. Sum over $\phi \in \Phi$ and we see the only ϕ 's making contributions are those in Φ_v , and we get the required equation. $\square$

Corollary 3.6. *With the above settings, for all primes $\mathfrak{p}_v$ of E dividing p ,*

$$\frac{ord_v(\pi)}{ord_v(q)} = \frac{\#\Phi_v}{\#H_v}$$

where the residue field $k = \mathcal{O}_K/\mathfrak{P} = \mathbb{F}_q$

Proof. p is unramified in E implies $ord_v(p) = 1$. Also $q = p^{f(\mathfrak{P}/p)}$ by definition. Therefore,

$$ord_v(q) = f(\mathfrak{P}/p)$$

On the other hand, corollary 3.7 gives

$$ord_v(\pi) = \sum_{\phi \in \Phi_v} f(\mathfrak{P}/\phi \mathfrak{p}_v)$$

and so

$$\frac{ord_v(\pi)}{ord_v(q)} = \sum_{\phi \in \Phi_v} \frac{1}{f(\phi \mathfrak{p}_v/p)} = \sum_{\phi \in \Phi_v} \frac{1}{f(\mathfrak{p}_v/p)} = \frac{\#\Phi_v}{\#H_v}$$

Here we use the fact that $f(\mathfrak{p}_v/p) = \#H_v$ because elements of H_v corresponds to the embeddings of the residue field of E w.r.t. $\mathfrak{p}_v$ into $\overline{\mathbb{F}_p}$ $\square$

Remark 3.2. Since abelian varieties with CM over a number field K have potentially good reduction everywhere, we may always pass to a finite extension K'/K that yields good reduction, so the obstacles to apply this formula are easy to bypass.

The similarity between corollary 3.6 and theorem 2.4 (c) essentially determines how we choose the CM type in the proof of Honda-Tate.

4 Honda-Tate Theorem

4.1 Proof of the Theorem

Honda-Tate theorem says that the map $[A] \mapsto [\pi_A]$ is bijective up to isogeny and conjugacy class for simple abelian varieties A over finite fields and Weil q -number π_A . By the corollary of Poincaré reducibility theorem, it gives a complete classification of abelian varieties over finite fields up to isogeny class. The toughest part is to show surjectivity, which was firstly shown in Honda's paper [Hon68]. The idea in this proof has deep influence in the study of lifting problems.

The fact that the map $[A] \mapsto [\pi_A]$ is well-defined and injective follows directly from the corollary of Tate's theorem. Indeed, if $A \sim B$, then $f_A = f_B$ by theorem 2.3. Also theorem 2.4 tells us that f_A and f_B are powers of the minimal polynomials of π_A and π_B respectively so they must agree. This shows that π_A and π_B are conjugates. If $\pi_A \sim \pi_B$ for A, B simple abelian varieties over $\mathbb{F}_q$, then they share the same minimal polynomial, say h . Then theorem 2.4 says that f_A and f_B are powers of h , so one must divide the other, say $f_A \mid f_B$. By the remark after theorem 2.3, this implies that A is isogenous to a subvariety of B , and hence A is isogenous to B since A, B are simple. This shows injectivity.

As noted, showing the surjectivity of this map is more interesting. We start with the definition of Weil restriction. Across this section, π is a given Weil q -number.

Let L/K be a finite extension of fields, and X a scheme over L . Consider the contravariant functor

$$Res_{L/K}X : (Sch/K)^{op} \rightarrow Sets, \quad S \mapsto Hom_L(S \times_K L, X) = X(S \times_K L)$$

For base field extension L/K , $Res_{L/K}X$ is always representable. For more general base change $h : S \rightarrow S'$, the result can found in [BLR90, theorem 7.6.4].

Definition 4.1. *The scheme over K representing $Res_{L/K}X$ is called the **Weil Restriction** of X w.r.t. L/K , which we will also denote by $Res_{L/K}X$. In particular, the K -rational points of $Res_{L/K}(X)$ are exactly the L -rational points of X*

Remark 4.1. Naturally one would ask what properties $Res_{L/K}X$ can inherit from X . This can be found in [BLR90, proposition 7.6.5]

Theorem 4.1. *Let k be a field and k' a finite extension of k . Let A be an abelian variety over k' and $\ell \neq \text{char}(k)$. Then, we have $T_\ell(Res_{k'/k}(A)) = Ind_{\Gamma_k}^{\Gamma_{k'}} T_\ell(A)$.*

Proof. By definition $Res_{k'/k}(A)(\bar{k}) = A(\bar{k} \otimes_k k')$. But $\bar{k} \otimes_k k' \cong \prod_{\sigma: k' \rightarrow \bar{k}} \bar{k}$, where σ is k -embedding of k' . Hence $Res_{k'/k}(A)(\bar{k}) = A(\prod_{\sigma} \bar{k}) = \prod_{\sigma} A(\bar{k})$. Take ℓ^n -torsion and pass to inverse limit, we get $T_\ell(Res_{k'/k}(A)) = \prod_{\sigma} T_\ell A$. Each σ extends uniquely to an element in Γ_k , and together they form a representative set of the quotient $\Gamma_k/\Gamma_{k'}$. So $T_\ell(Res_{k'/k}(A)) = Ind_{\Gamma_{k'}}^{\Gamma_k} T_\ell(A)$ by definition. $\square$

Lemma 4.1. *Let k' be a extension field of $k = \mathbb{F}_q$ of degree n , and let A' be a simple abelian variety over k' . If $A = Res_{k'/k}(A')$ is the Weil restriction of A' to k then $f_A(x) = f_{A'}(x^n)$. In particular, the Weil- q -numbers $\pi_{A'}$ and π_A^n are conjugate.*

Proof. This follows from theorem 4.1 immediately. $\square$

Corollary 4.1. *If π is a Weil- q -number such that π^n is conjugate to the Frobenius of a simple abelian variety over $k' = \mathbb{F}_{q^n}$, then π is conjugate to the Frobenius of a simple abelian variety over $k = \mathbb{F}_q$.*

Proof. If A' is a simple abelian variety over $\mathbb{F}_{q^n}$ whose Frobenius is conjugate to π^n , then π^n is a root of $f_{A'}$. By lemma 4.1, π is therefore a root of f_A where $A = Res_{k'/k}(A')$. By the observation in theorem 1.6 ($f_A = \prod_{i=1}^n f_{A_i}$ where each simple abelian variety A_i occurs in the decomposition of A), π is conjugate to the Frobenius of a simple abelian variety over $\mathbb{F}_q$. $\square$

Corollary 4.1 tells us that if we can find a power of π that has a lift (which is called "effective" in some notes) by $A \mapsto [\pi_A]$, then π has a lift as well. This means that we can freely replace π with its power if necessary.

The next step is to construct an abelian variety with CM by a very specified CM-type which by the Shimura-Taniyama formula determines how the Frobenius behaves as an element in L for A of CM by L . It ensures that the Frobenius is conjugate to a power of π by the following lemma

Lemma 4.2. *If π, π' are a Weil q and q' -numbers, respectively, with $\mathbb{Q}(\pi), \mathbb{Q}(\pi') \subseteq L$ such that $\frac{\text{ord}_w(\pi')}{\text{ord}_w(q')} = \frac{\text{ord}_w(\pi)}{\text{ord}_w(q)}$ for all places $w \mid p$ of L , then there exists $N, N' \in \mathbb{Z} \geq 1$ such that $\pi^N = (\pi')^{N'}$.*

Proof. After replacing π and π' by suitable powers we may assume that $q = q'$. So $\text{ord}_w(\pi/\pi') = 0$, i.e. $|\pi/\pi'|_w = 1$ for $w \mid p$. At other finite places π and π' are units since they both divide q . At the infinite place π and π' both have absolute value $q^{\frac{1}{2}}$. So π/π' has absolute value 1 at each place of L , which implies that it is a root of unity. Hence some power of π/π' is 1, and the lemma follows. $\square$

As hinted after corollary 3.6, the CM type (L, Φ) will be built around the value $\frac{\text{ord}_v(\pi)}{\text{ord}_v(q)}$ for each $v \mid p$ with the help of Brauer groups. Recall in class field theory, we have the **invariant map** from the Brauer group of k

$$\text{inv} : Br(k) \hookrightarrow \mathbb{Q}/\mathbb{Z}$$

by taking the Galois cohomology of the Kummer sequence and passing to the colimit. It is an isomorphism when k is a local field. If $k = \mathbb{R}$, then the image is $\frac{1}{2}\mathbb{Z}/\mathbb{Z}$, and if $k = \mathbb{C}$, then $Br(k) = 0$. (See [Ser79])

Moreover, we have a short exact sequence that relates the local and global Brauer groups

$$0 \rightarrow Br(k) \rightarrow \oplus_v Br(k_v) \xrightarrow{(\text{inv})_v} \mathbb{Q}/\mathbb{Z} \rightarrow 0 \quad (4.1)$$

Here the sum is over all places of k . The first map sends $[D]$ to $([D \otimes_k k_v])_v$. The second maps sends $(a_v)_v$ to $\sum \text{inv}_v(a_v)$.

Lemma 4.3. *For π a Weil q -number, $\mathbb{Q}(\pi)$ is either totally real or a CM field.*

Proof. If π is real, then $\pi = \pm q^{1/2}$ and it follows.

If π is not real, then for each conjugate ρ of π , $|\rho| = q^{1/2}$ and hence $\rho + q/\rho$ is real. So the conjugates of $\pi + q/\pi$ are all real and hence $\mathbb{Q}(\pi + q/\pi)$ is totally real, over which $\mathbb{Q}(\pi)$ is of degree 2. So $\mathbb{Q}(\pi)$ is a CM field. $\square$

Theorem 4.2. *There exists a division algebra $D/\mathbb{Q}(\pi)$ satisfying local invariant conditions as in theorem 2.4 (c). In particular, let $d = [D : \mathbb{Q}(\pi)]^{1/2}$ and $e = [\mathbb{Q}(\pi) : \mathbb{Q}]$, then $d \cdot e = 2g$ for some g .*

Proof. By the exactness of (4.1) and the fact that local invariants are isomorphisms for local fields, such D exists and is unique up to isomorphism if and only if $\sum \text{inv}_v = 0 \pmod{1}$ with the values in theorem 2.4, i.e. $\text{inv}_v(D) = \frac{\text{ord}_v(\pi)}{\text{ord}_v(q)} [K_v : \mathbb{Q}_p]$ if $v \mid p$, $\text{inv}_v = \frac{1}{2}$ if v is real, and 0 otherwise. By lemma 4.3, $\mathbb{Q}(\pi)$ is either totally real or a CM field. If $\mathbb{Q}(\pi)$ is totally real, then $\pi = \pm q^{1/2}$ and $\mathbb{Q}(\pi) = \mathbb{Q}$ or $\mathbb{Q}(p^{1/2})$. In the first case, $\text{inv}_p = 1/2 = \text{inv}_\infty$. Now if $\mathbb{Q}(\pi) = \mathbb{Q}(p^{1/2})$, p is totally ramified at $p^{1/2}$. It easy to compute that $\text{inv}_{p^{1/2}} = 1 = 0 \pmod{1}$. $\mathbb{Q}(p^{1/2})$ has two real embeddings, namely identity and the one maps $p^{1/2}$ to $-p^{1/2}$. So both cases follow.

If $\mathbb{Q}(\pi)$ is a CM field, we have $\pi \cdot \bar{\pi} = q$. Since there are no real places, the only possible non-trivial contribution to $\sum inv_v$ is from $v \mid p$. Then

$$inv_v + inv_{\bar{v}} = \frac{ord_v(\pi) + ord_v(\bar{\pi})}{ord_v(q)} [K_v : \mathbb{Q}_p] = 0 \in \mathbb{Q}/\mathbb{Z}$$

Since each v and $\bar{v}$ come in pairs, $\sum inv_v = 0$ □

Proposition 4.1. *Given D from theorem 4.2, there exists a CM field $L \supseteq \mathbb{Q}(\pi) = F$ such that $D \otimes_F L$ splits (i.e. $inv_v(D \otimes_F L) = 0$) at all places of L . And we have $[L : F] = [D : F]^{1/2} = d$ and hence L is of degree $2g$.*

Proof. If F is totally real, we can take $L = F(\sqrt{-q})$. D splits over L as the invariants are multiplied by 2.

If F is a CM field, we have seen that $F^+ = \mathbb{Q}(\pi + q/\pi)$ is its maximally totally real subfield. Then there exists a totally real extension L^+/F^+ of degree d such that for all places v of F^+ over p stay inert. In particular, this means that $[L_w^+ : F_v^+] = d$ for each unique $w \mid v$. Then the compositum $L = L^+ F$ is a CM field, and $[L : F] = [L^+ : F^+] = d$.

Note that $inv_w(D \otimes_F L) = inv_v(D)[L_w : F_v] = inv_v(D) \cdot d$ for $w \mid v$. Then $inv_w(D \otimes_F L) = 0$ for w above $v \nmid p$. In the case $v \mid p$, $inv_v(D)$ is a multiple of $1/d$ since D is of degree d over F and hence $inv_w(D \otimes_F L) = 0 \pmod{1}$ when $w \mid p$. And there are no real places since L and F are both CM fields. Hence D splits over L . □

To see the magic of the CM field L that arises from proposition 4.1, we now choose a CM type $\Phi \subseteq Hom(L, \mathbb{C})$, which amounts to choose $\Phi_w \subseteq H_w$ for each place w of L . We want the resulting Φ satisfies $\frac{\#\Phi_w}{\#H_w} = \frac{ord_v(\pi)}{ord_v(q)}$ as implied in corollary 3.6 for all $w \mid p$ of L . This is possible because

$$\#H_w \cdot \frac{ord_v(\pi)}{ord_v(q)} = [L_w : \mathbb{Q}_p] \frac{ord_v(\pi)}{ord_v(q)} = [L_w : F_v] inv_v(D) = inv_w(D \otimes_F L) \in \mathbb{Z}$$

because by definition $H_w = Hom(L_w, \overline{\mathbb{Q}_p})$ and the rest is by our choices of D and L .

We also have $\#\Phi_w + \#\Phi_{\bar{w}} = \#H_w = \#H_{\bar{w}}$ because $\pi \cdot \bar{\pi} = q$. This means that in our choice $\overline{\Phi_w}$ is the complement of $\Phi_{\bar{w}}$ in $H_{\bar{w}}$. Hence $\Phi = \bigsqcup \Phi_w$ is a CM-type.

We are ready to build an abelian variety of CM with (L, Φ) given above. The idea is to first construct such abelian variety over $\mathbb{C}$ and then descend to a number field with CM structure unchanged. By proposition 3.3, it has good reduction after finite extension. Then the special fibre of the integral model is almost what we want.

Lemma 4.4. *Given L as above and a CM-type (L, Φ) , there is an abelian variety A of dimension g over $\mathbb{C}$ of the given CM-type.*

Sketch of proof. Let $\mathbb{C}_\Phi = \prod_{\phi \in \Phi} \mathbb{C}$. The natural inclusion $L \hookrightarrow \mathbb{C}_\Phi$ identifies $\mathcal{O}_L$ with a lattice of $\mathbb{C}_\Phi$. One can show that the Riemann relations are always satisfied for a lattice arising in this way, and hence the complex torus $\mathbb{C}_\Phi/\mathcal{O}_L$ is an abelian variety of the given CM-type. □

Lemma 4.5. *The resulting abelian variety $A/\mathbb{C}$ in lemma 4.4 can descend to a number field K with the same CM structure.*

Sketch of proof. First we descend A to $\overline{\mathbb{Q}}$. Note that $\mathbb{C}$ is a colimit of its finitely-generated $\overline{\mathbb{Q}}$ -subalgebras. Since A is finitely presented, A can descend to an abelian scheme over a finitely-generated $\overline{\mathbb{Q}}$ -algebra R . We may assume that the tangent space is now isomorphic to $\prod_{\phi \in \Phi} R_{\phi}$ as a free $L \otimes_{\mathbb{Q}} R$ module, i.e. the CM type is unaffected.

Let m be a maximal ideal of R . The fibre at m is then an abelian variety over $R/m = \overline{\mathbb{Q}}$ of CM type (L, Φ) . Since $\overline{\mathbb{Q}}$ is the colimit of number fields, the same argument shows that it can further descend to one of them. $\square$

And finally,

Proposition 4.2. *There exists an abelian scheme $\mathcal{A}$ over $\mathcal{O}_v$, where $v \mid p$ is a valuation of a number field K , such that $\mathcal{A}_K$ admits CM by L from Proposition 4.1. Consequently, $\mathcal{A}_k$ has Frobenius conjugate to π^N for some $N \in \mathbb{Z}$.*

Proof. Lemma 4.5 gives an abelian variety A of dimension g with CM over a number field K of CM-type (L, Φ) . By proposition 3.3, after a finite extension over K we may assume A has good reduction at $v \mid p$. Then its Néron model $\mathcal{A}/\mathcal{O}_v$ is an abelian scheme, and its special fibre $\mathcal{A}_k$ is an abelian variety over $k = \mathbb{F}_{q_0}$ for some p -power q_0 .

By lemma 3.3, $L \hookrightarrow \text{End}^0(A) \cong \text{End}^0(\mathcal{A}) \hookrightarrow \text{End}^0(\mathcal{A}_k)$ with $\dim L = 2\dim \mathcal{A}_K$, and hence $\mathcal{A}_K$ admits CM by L . Moreover, the Frobenius $\pi_0 \in \mathcal{O}_L$ of $\mathcal{A}_k$ satisfies corollary 3.6 of the Shimura-Taniyama formula. That is,

$$\frac{\text{ord}_v(\pi_0)}{\text{ord}_v(q_0)} = \frac{\#\Phi_v}{\#H_v} = \frac{\text{ord}_v(\pi)}{\text{ord}_v(q)}$$

for all $v \mid p$. By lemma 4.2 we conclude that $\pi^N = \pi_0^{N_0}$ for some integers N, N_0 , and we may replace L with a degree N_0 totally ramified extension to get $\pi_0 = \pi^N$ $\square$

Now by corollary 4.1, we just showed the surjectivity of the map $[A] \mapsto [\pi_A]$, which completes the proof of Honda-Tate theorem.

4.2 More on CM Lifting

Building on earlier work of Honda, Tate gave an improved lifting theorem in [Tat71] which initiates the study of CM liftings of abelian variety over finite fields.

Theorem 4.3. *For any isotypic abelian variety A over a finite field k , after a finite extension k' over k $A_{k'}$ is isogenous to the reduction of a CM abelian variety with good reduction over a p -adic field with residue field k' , i.e. $A_{k'}$ admits a CM lifting to this p -adic field.*

Several decades later, people refined this result by removing the requirement of fields extension and gave a modern version of the corollary of Honda-Tate:

Theorem 4.4. *(Ching-Li Chai-Brian Conrad-Frans Oort) Every abelian variety A over a finite field k is isogenous to an abelian variety A' that has a CM lifting over a mixed characteristic local domain with residue field k .*

In the proof, a key step is to give a CM lifting of the associated **p-divisible group**, then the **Serre-Tate theorem** tells us that the original abelian variety admits a formal CM lifting to a formal abelian scheme over some local algebra. More details can be found in [CCO14].

4.3 Manin's Problem

In [Tat71], Tate also applied Honda-Tate to solve a conjecture on Newton polygon by Manin in 1963

Definition 4.2. Let A be an abelian variety over $\mathbb{F}_{p^n}$ of dimension g . Recall that f_A is the minimal polynomial of π_A acting on $V_\ell A$. Then by proposition 1.7 $f_A \in \mathbb{Z}[x]$. Let $f_A = \sum a_i x^{2g-i}$. The **Newton polygon** $\mathcal{N}(A)$ of A is the lower convex hull of the set of points $\{(i, v_p(a_i)/n) | 0 \leq i \leq 2g\}$, which is the Newton polygon of f_A compressed by size n

Newton polygons can also be described by its slopes $0 \leq d_1 \leq \dots \leq d_{2g} \leq 1$, which can be represented by data $\sum_i (b_i, c_i)$ where $b_i, c_i \in \mathbb{Z}$, $\gcd(b_i, c_i) = 1$ and $b_i/(b_i + c_i) \leq b_{i+1}/(b_{i+1} + c_{i+1})$ and $\sum_i (b_i + c_i) = 2g$. We can recover the Newton polygon from these data by setting the slopes $b_i/(b_i + c_i)$ with multiplicities $b_i + c_i$ in ascending order.

We say a Newton polygon ξ **symmetric** if the slopes satisfy that $d_i + d_{1+2g-i} = 1$. When in the form of the sum of ordered pairs, this is equivalent to say that $(b_i, c_i) \in \xi$ if and only if $(c_i, b_i) \in \xi$

Example 2. For a supersingular elliptic curves E over $\mathbb{F}_p$, $f_E = x^2 + ax + p$ where $p \mid a$. Hence its Newton polygon is the straight line $(0, 0)$ to $(2, 1)$, which can be represented by the ordered pair $(1, 1)$. If E is ordinary, then $p \nmid a$ and so its Newton polygon is generated by $(0, 0)$, $(1, 0)$ and $(2, 1)$, and can be represented by the data $(0, 1) + (1, 0)$

Theorem 4.5. (Manin's problem) Suppose given a prime number p and a symmetric Newton polygon ξ . Then there exists an abelian variety A over a field of characteristic p with $\mathcal{N}(A) = \xi$

Sketch of Proof. A symmetric Newton polygon can be written as

$$\xi = a \cdot (1, 1) + b \cdot ((1, 0) + (0, 1)) + \sum_i ((d_i, c_i) + (c_i, d_i))$$

where $a, b \geq 0$ and $d_i > c_i > 0$ coprime.

Note $\mathcal{N}(A) \cup \mathcal{N}(B) = \mathcal{N}(A \times B)$, where $\mathcal{N}(A) \cup \mathcal{N}(B)$ denotes the Newton polygon obtained by taking all slopes in $\mathcal{N}(A)$ and $\mathcal{N}(B)$ and reordering in ascending order. In example 2 we already showed that a supersingular elliptic curve E has $\mathcal{N}(E) = (1, 1)$, while for ordinary elliptic curve E' we have $\mathcal{N}(E') = (1, 0) + (0, 1)$. So it suffices to find an abelian variety A with $\mathcal{N}(A) = (d, c) + (c, d)$ for every $d > c > 0$ coprime.

Let $n = c + d$ and $q = p^n$. Consider the polynomial $P(x) = x^2 + p^c x + p^n$ and its root π . Then π is a Wéil q -number, and Honda-Tate gives a simple abelian variety A of dimension g over $\mathbb{F}_q$ with $\pi_A = \pi$ (we replace π with $\bar{\pi}$ if necessary). Let $D = \text{End}^0(A)$. Theorem 2.4 tells us that D is a central division algebra over the CM field $F = \mathbb{Q}(\pi)$.

Note that $P(x/p^c) = (x/p^c)^2 + (x/p^c) + p^{d-c}$, and this implies that p splits completely into two places $v, \bar{v}$ in F . We can also observe that the p -adic value of π is either c or d . By theorem 2.4 (c), the invariants of D/F at v and $\bar{v}$ are c/n and d/n . Since $\gcd(c, n) = \gcd(c, d) = 1$, we have $[D : F] = n^2$ and theorem 2.4 (b) gives $g = [D : F]^{1/2} [F : \mathbb{Q}] / 2 = n$. So A is of dimension n .

Now [Tat71] §2 shows that the p -divisible group $A[p^\infty]$ is isogenous to the group $G_{d,c} \times G_{c,d}$ from Dieudonné theory, where $G_{d,c}$ is the unique isoclinic (only one slope occurs in the Newton polygon) p -divisible group of slope $d/(d+c) = d/n$. Consequently, $G_{d,c} \times G_{c,d}$ is algebraizable, which produces an abelian variety with Newton polygon $(d, c) + (c, d)$. So we are done. $\square$

Remark 4.2. The notations in this section follow those in Frans Oort's paper [Oor00], where he gives a different approach to theorem 4.5 with modern techniques.

5 p-Divisible Groups

5.1 Finite Flat Group Schemes

Honda's work gives the criterion regarding the existence of CM lifting of abelian varieties over finite fields. In the following two chapters we give an alternative approach for more general liftings using p -divisible groups, which is the content of the Serre-Tate theorem

The main ingredient of p -divisible groups is finite flat group schemes.

Definition 5.1. *Let S be a scheme and G/S be a group scheme. We call G/S a **finite flat group schemes** of order n if the structure morphism $G \rightarrow S$ is locally free of rank n . In the affine case, $G = \text{Spec}(A)$ over $S = \text{Spec}(R)$ is finite flat if A is a finitely generated flat R -module.*

Finite flat group schemes naturally arise when we consider the kernel of isogenies between abelian schemes. For example, let A/S be an abelian scheme of dimension g . Then, for any $n \in \mathbb{N}$, the group scheme $A[n] = \ker([n] : A \rightarrow A)$ over S is finite flat of order n^{2g}

Recall that the **Cartier Dual** of a finite flat commutative group scheme G over S is the finite flat commutative group scheme $G^\vee$ representing the functor $T \mapsto \text{Hom}_{gp}(G_T, \mathbb{G}_{m,T})$ for scheme T over S . Any S -homomorphism $G \rightarrow G'$ induces a dual map $G'^\vee \rightarrow G^\vee$ by pre-composition. If $G = \text{Spec}(A)$ over $S = \text{Spec}(R)$ is affine, then $A^\vee = \text{Hom}_R(A, R)$ is a Hopf R -algebra and so $G^\vee = \text{Spec}(A^\vee)$ is a finite flat group scheme over S with the same order as G . Moreover, there is a canonical isomorphism $(G \times H)^\vee \cong G^\vee \times H^\vee$.

Theorem 5.1. (Deligne) *Let G/S be a finite flat commutative group scheme of order n . Then $[n]$ annihilates G , i.e. the map $[n] : G \rightarrow G$ factors through the unit section $e : S \rightarrow G$*

To prove this, Deligne defined the so-called trace map. Here we follow the notations in [TO70]. Suppose G, S are affine as above and $T = \text{Spec}(B)$ is any S -scheme with structure map $f : T \rightarrow S$. Then Tr_f is the unique map to make the following diagram commute:

$$\begin{array}{ccc} G(T) & \hookrightarrow & A^\vee \otimes B \\ \downarrow \text{Tr}_f & & \downarrow N \\ G(S) & \hookrightarrow & A^\vee \end{array}$$

where the horizontal arrows are the easily seen inclusions and vertical arrows are the norm maps. From the definition we can deduce that $\text{Tr}_f(f^*u) = u^n$ for any $u \in G(S)$ and $\text{Tr}_f(v) = \text{Tr}_f(v \circ t)$ for any $v \in G(T)$ and $t : T \rightarrow T$ any S -automorphism.

Now it suffices to show that for any $u \in G(S)$, $[n]u = u^n = 1$. We take $T = G$ and consider $\text{id}_A \in G(G) = \text{Hom}_R(A, A)$. Let $t_u = G \cong G \times_S S \xrightarrow{(1,u)} G \times_S G \rightarrow G$ be the translation on G by u . Then $\text{id}_A \circ t_u = f^*u \cdot \text{id}_A$ in $G(G)$, and so

$$\text{Tr}_f(\text{id}_A) = \text{Tr}_f(f^*u \cdot \text{id}_A) = \text{Tr}_f(f^*(u))\text{Tr}_f(\text{id}_A) = u^n \text{Tr}_f(\text{id}_A)$$

in $G(S)$. We conclude that $u^n = 1$ since $\text{Tr}_f(\text{id}_A)$ is invertible.

Remark 5.1. Details of explicit calculations can be found in [TO70]. It is still unknown whether the theorem holds when G is non-commutative.

Definition 5.2. *A local homomorphism $f : A \rightarrow B$ of local rings is **unramified** if $f(m_A)B = m_B$ and the field B/m_B is finite and separable over A/m_A . A morphism $\varphi : X \rightarrow Y$ of schemes is **unramified** at $x \in X$ if it is of finite type and the local homomorphism $\mathcal{O}_{Y,\varphi(x)} \rightarrow \mathcal{O}_{X,x}$ is unramified. We say φ is unramified if it is unramified all over X .*

Note that this agrees with the definition in algebraic number theory where we only consider DVR. The unramification of a morphism of finite type can be detected by its modules of relative differentials $\Omega_{X/Y}$ which vanishes if φ is unramified and vice versa. The equivalence of these conditions is proved in [BLR90, section 2.2].

Following this, things are even easier for finite flat group schemes owing to their group structure

Definition 5.3. The *augmentation ideal* I of a group scheme $G = \text{Spec}(A)$ over $S = \text{Spec}(R)$ is kernel of the counit $A \rightarrow R$.

Lemma 5.1. Let $G = \text{Spec}(A)$ be a finite flat group scheme over $S = \text{Spec}(R)$ with augmentation ideal I . Then there exist natural isomorphisms

$$I/I^2 \otimes_R A \cong \Omega_{A/R} \quad \text{and} \quad I/I^2 \cong \Omega_{A/R} \otimes_A A/I$$

(notice that I/I^2 is naturally an A/I -module.) Consequently, G is unramified (and hence étale) if and only if $I = I^2$.

Indeed, the second isomorphism follows from the canonical conormal exact sequence

$$0 \rightarrow I/I^2 \rightarrow \Omega_{A/R} \otimes_A A/I \rightarrow \Omega_{R/R} \rightarrow 0$$

since $\Omega_{R/R}$ vanishes.

Definition 5.4. Let S be a scheme and G/S be a group scheme. We call G/S *étale* if the structure map $G \rightarrow S$ is étale, i.e. it is flat, unramified and locally of finite presentation.

Example 3. Constant group schemes are étale. Indeed, let M be a finite abelian group and $\underline{M} \cong \text{Spec}(\prod_{i \in M} R)$ its constant group schemes over $S = \text{Spec}(R)$ with unit section given by the projection to the factor for $i = 0$. The augmentation ideal I of $\underline{M}$ is $\prod_{i \in M, i \neq 0} R$, and hence $I = I^2$. So $\underline{M}$ is unramified and hence étale over S .

Étale morphisms are analogs of local isomorphisms of manifolds in differential geometry. We will see that it can also be characterized by the infinitesimal lifting property. The next theorem shows that étale k -group schemes are controlled by their geometric points.

Theorem 5.2. There is an equivalence of categories between the category of finite étale k -group schemes and the category of finite Γ_k -groups by $G \mapsto G(\bar{k})$. Moreover, G and $G(\bar{k})$ have the same order.

Corollary 5.1. Let G/S be a finite flat group scheme of order n . If n is invertible in S , then G/S is étale.

Proof. It suffices to show that $\Omega_{G/S} = 0$. By Deligne's theorem, we can factorize the map $[n]$ as

$$[n] : G \rightarrow S \xrightarrow{e} G$$

It induces

$$\Omega_{G/S} \rightarrow \Omega_{S/S} \rightarrow \Omega_{G/S}$$

and thus is the zero map. Also note that for each geometric point s of S , the map

$$(\Omega_{G/S})_s \rightarrow (\Omega_{G/S})_s$$

is just multiplication by n which is an isomorphism by our assumption. This implies that $\Omega_{G/S} = 0$ as desired. $\square$

It follows immediately that the n -torsion finite flat group scheme $A[n]$ of an abelian variety A over k with $\text{char}(k) = p$ is étale if p does not divide n . On the other hand, as noted before, $A[p]$ is never étale, and we need more general techniques on finite flat group schemes to deal with them

Theorem 5.3. (*The connected-étale sequence*) *Let G be a finite flat group scheme over a field k and G^0 be the connected component of G containing the unit section then $G^{et} = G/G^0$ is étale, i.e. there is a short exact sequence*

$$0 \rightarrow G^0 \rightarrow G \rightarrow G^{et} \rightarrow 0$$

Moreover, the short exact sequence splits if k is perfect.

We first need to understand G^0 and G^{et} . For $G = \text{Spec}(A)$ with A finite flat over k , it is known that $A \cong \prod_i A_i$ where A_i 's are local k -algebras which correspond to its connected components. It is easily seen that the unit section $A \cong \prod_i A_i \rightarrow k$ has all but one $e_i \mapsto 0$ since there's no zero-divisors in the image, and WLOG we assume it is e_0 . So we have $A \rightarrow A_0 \rightarrow k$. Then $G^0 = \text{Spec}(A_0)$ is the connected component of the identity.

Similarly we let A_{et} be the maximal étale subalgebra of A . It can be shown that $A_{et} = \prod_i k_i^{sep}$ and is a Hopf algebra where k_i is the residue field of A_i and k_i^{sep} is the separable closure of k in k_i . Notice that $G^{et} = \text{Spec}(A_{et}) = \bigsqcup \text{Spec}(k_i^{sep})$ is merely the collection of isolated points in G .

We want to show $G^0 \cong \ker(G \rightarrow G^{et})$ where $G \rightarrow G^{et}$ is induced by $A_{et} \hookrightarrow A$. Note that the unit section $A_{et} \rightarrow k$ of G^{et} factors through k_0 by our choice of A_0 . So the augmentation ideal in G_{et} is $\prod_{i \neq 0} k_i^{sep}$. Hence $\ker(G \rightarrow G^{et})$ is represented by $A/(\prod_{i \neq 0} k_i^{sep})A$, which is isomorphic to A_0 since $A \cong \prod_i A_i$.

Remark 5.2. One can generalize this result to group schemes over a noetherian complete local ring.

Since we only consider k being perfect, we have the decomposition $A[p] = A[p]^o \times A[p]^{et}$ for the p -torsion of A over k . One of our remaining tasks is to determine the possibilities here using Dieudonné theory, but first we introduce the definition of p -divisible groups.

5.2 p -Divisible Groups

Note that ℓ -adic Tate modules only concern the geometric points on abelian varieties. For prime $\ell \neq \text{char}(K)$, corollary 5.1 shows that $A[\ell^n]$'s are étale and hence controlled by their geometric points as in theorem 5.2, so it is enough to study ℓ -adic Tate modules.

On the other hand, Tate module is less efficient when $\ell = p = \text{char}(k)$ as $A[p]$'s connected part completely collapses to the identity when taking geometric points. In the most extreme case, for a supersingular elliptic curve E , $E[p](\bar{k}) = 0$ and so $T_p(E) = 0$. So its p -adic Tate module provides us nothing on its p -torsion structure!

To fix this, Tate instead considered the colimit of p -power torsion subgroup schemes of A , which naturally motivates the most general definition of p -divisible groups.

Definition 5.5. *A p -divisible group of height h over a scheme S is an inductive system $(G_n, i_n)_{n \in \mathbb{N}}$ with each G_n finite flat commutative of order p^{nh} and i_n closed immersions such that G_n can be identified with the kernel of the map $G_{n+1} \xrightarrow{[p^n]} G_{n+1}$ via i_n for $n \geq 1$. Without ambiguity we will also use p -divisible groups to refer their colimits $G = \varinjlim G_n$,*

Remark 5.3. Let $i_{v,u} : G_v \rightarrow G_{v+u}$ denote $i_{v+u-1} \circ \dots \circ i_v$, i.e. iterating the closed immersions u times starting from G_v . A diagram chase shows that $G_{v+u} \xrightarrow{p^u} G_{v+u}$ factors through $i_{v,u}$ via a map $j_{u,v} : G_{v+u} \rightarrow G_v$, i.e. $p^u = i_{v,u} \circ j_{u,v}$ and furthermore, we have a short exact sequence

$$0 \rightarrow G_v \xrightarrow{i_{v,u}} G_{v+u} \xrightarrow{j_{u,v}} G_u \rightarrow 0$$

by considering their orders.

Corollary 5.2. $[p] : G \rightarrow G$ is surjective for a p -divisible group G .

Indeed, in the colimit $G = \varinjlim G_n$ we regard each closed immersion i_n as an inclusion and the induced map $j_{u,v}$ as $[p^v]$. So this follows from remark 5.3 immediately.

Remark 5.4. It is clear from the definition that $(A[p^n], i)_{n \in \mathbb{N}}$ is a p -divisible group with inclusions. We will denote it (and its colimit) by $A[p^\infty]$. If A has dimension g , then $A[p^\infty]$ is of height $2g$.

Example 4. Here are two most important examples of p -divisible groups over S :

1. The system of constant group schemes $(\mathbb{Z}/p^n\mathbb{Z}, i_n)$ of height 1, where $i_n : \mathbb{Z}/p^n\mathbb{Z} \rightarrow \mathbb{Z}/p^{n+1}\mathbb{Z}$ is the natural inclusion using multiplication by p . We denote it by $\mathbb{Q}_p/\mathbb{Z}_p$.
2. The height-1 p -divisible group of the multiplicative group $\mathbb{G}_m$ over S , i.e. $(\mathbb{G}_m[p^n] = \mu_{p^n}, i_n)$ where i_n is the natural inclusion. We denote it by μ_{p^∞} .

We say a p -divisible group G is étale if each G_n is étale, or connected if each G_n is connected. Then $\mathbb{Q}_p/\mathbb{Z}_p$ is étale over S because the constant group schemes are étale.

The connected-étale sequence for finite flat group schemes over a perfect field k also exists for p -divisible groups, i.e. $G = G^o \times G^{et}$ for a p -divisible group G over k , where $G^o = \varinjlim G_n^o$ and $G^{et} = \varinjlim G_n^{et}$. (G_n^o) and (G_n^{et}) are p -divisible groups can be seen by the facts that $G_{n+1}/G_n \cong G_1$ from the short exact sequence in remark 5.3, and that $(-)^o$ and $(-)^{et}$ are both exact functors.

The moral is that G_1 can determine G to some extent. This gives us much convenience because, due to the limitation of its order, the structure of G_1 is usually very clear.

By computing the maps of their Hopf algebra structure, we see that $\mathbb{Z}/p^n\mathbb{Z}$ is the Cartier dual of μ_{p^n} for each n . In general, for $G = \varinjlim G_n$, if $G^\vee = \varinjlim G_n^\vee$ with the transition map induce by $[p]_G$ is a p -divisible group, we refer to it as the Cartier dual of G as p -divisible groups. In particular, $\mathbb{Q}_p/\mathbb{Z}_p = (\mu_{p^\infty})^\vee$.

5.3 Dieudonné Theory

Dieudonné theory was first developed to study formal Lie groups which we will see have deep connections with p -divisible groups. Today it has been heavily applied in both representation theory of finite group schemes and p -adic hodge theory. Our main goal is to see its application in classifying finite flat group schemes of p -power order and p -divisible groups order over a perfect field k with characteristic p .

Let G be a finite commutative group scheme over $S = \text{Spec}(k)$. Recall that for S -scheme G there is the **absolute Frobenius** $F_G : G \rightarrow G$. If $G = \text{Spec}(A)$, this is induced by $A \rightarrow A, a \mapsto a^p$. It also gives a commutative diagram

$$\begin{array}{ccc} G & \xrightarrow{F_G} & G \\ \downarrow & & \downarrow \\ S & \xrightarrow{F_S} & S \end{array}.$$

Definition 5.6. Recall that the twist of G by the absolute Frobenius of S is denoted by $G^{(p)}$. Then this diagram induces a unique morphism $F_{G/S} : G \rightarrow G^{(p)}$ called the **(relative) Frobenius** of G . We can also do this to $G^\vee$ and get a morphism $G^\vee \rightarrow (G^\vee)^{(p)}$. Then we dualize this map and use the fact that $(G^\vee)^{(p)} = (G^{(p)})^\vee$ to get a morphism $V_{G/S} : G^{(p)} \rightarrow G$, called **(relative) Verschiebung** of G .

Remark 5.5. Relative Frobenius and Verschiebung satisfy $F_{G/S} \circ V_{G/S} = [p]_{G^{(p)}}$, $V_{G/S} \circ F_{G/S} = [p]_G$.

Let $W = W(k)$ denotes the **ring of Witt vectors** of k and is a complete DVR of mixed characteristic with residue field k (see [Ser79]). If $k = \mathbb{F}_{p^n}$, then $W(k)$ is the ring of integers of the unique unramified extension (we denote it by $W(k)[1/p]$ in this case) of $\mathbb{Q}_p$ of degree n . Let σ denote the unique automorphism of $W(k)$ lifting the absolute Frobenius $x \mapsto x^p$ on k , i.e. $\sigma : \sum_{n=0}^{\infty} [a_n] p^n \mapsto \sum_{n=0}^{\infty} [a_n^p] p^n$.

To study a group scheme G over k , we can study its connected component G^o and étale part G^{et} separately. G^{et} is much easier to deal with, because it is controlled by the geometric points which form a Γ_k -group. This fails for the connected part (or local-type part in some books). However, people have found out that we can lift them to linear objects over W that preserve the action of Frobenius and Verschiebung. This leads to the definition of Dieudonné modules.

Definition 5.7. A **Dieudonné module** over k a W -module D together with two additive morphisms $F, V : D \rightarrow D$ such that $F(ad) = \sigma(a)F(d)$, $V(ad) = \sigma^{-1}(a)V(d)$ and $FV = VF = p$ for all $a \in W$ and $d \in D$.

We say F is σ -linear and V is σ^{-1} -linear. Note that a Dieudonné module can also be understood as a module over the ring $W[F, V]$ (noncommutative when $k \neq \mathbb{F}_p$) subject to the relations $FV = VF = p$, $Fc = \sigma(c)F$ and $Vc = \sigma^{-1}(c)V$ for all $c \in W$. In some notes this is called the **Dieudonné ring**.

The main theorem of Dieudonné theory tells us that we can understand commutative group schemes of p -power order in terms of Dieudonné modules.

Theorem 5.4. *There is an exact anti-equivalence M of categories between the category of finite commutative group schemes of p -power order over k and Dieudonné modules of finite length over W , with the following properties:*

1. M is an exact functor
2. $\#|G| = p^{\ell_W(M(G))}$
3. G is connected if and only if F is nilpotent on $M(G)$
4. G is étale if and only if F is an isomorphism on $M(G)$
5. $M(G)^\vee = M(G^\vee)$

Remark 5.6. By taking the colimit (and limit for the latter category), we get an exact anti-equivalence M between categories of p -divisible groups and Dieudonné modules which are free $W(k)$ -modules and $\text{rank}(M(G))$ agrees with the height of G .

We want to show how $M(G)$ is constructed for G a finite commutative group scheme of p -power order but first we need to give a more precise decomposition of G using Cartier duality. Recall that over a perfect field k we already have the decomposition $G = G^o \times G^{et}$. Using the property of Cartier duality, $G^o = ((G^o)^\vee)^\vee = (((G^o)^\vee)^o)^\vee \times (((G^o)^\vee)^{et})^\vee$, and similarly for G^{et} . Eventually we decompose G into 4 parts: $G = G_{ll} \times G_{le} \times G_{el} \times G_{ee}$, where G_{ll} is the connected part with connected dual, G_{le} is the connected part with étale dual and so on.

Remark 5.7. It is clear that $G^o = G_{ll} \times G_{le}$ and $G^{et} = G_{el} \times G_{ee}$, while for the multiplicative Jordan decomposition $G \cong G^{mu} \times G^{un}$, where the multiplicative part $G^{mu} = G_{le} \times G_{ee}$ and the unipotent part $G^{un} = G_{ll} \times G_{el}$ and we recollect that G^{mu} has étale dual and G^{un} has connected dual.

However, in our settings $G_{ee} = 0$ due to the next lemma

Lemma 5.2. *Let k be a perfect field with characteristic p and G a finite commutative group scheme over k of p -power order. If G is étale with étale dual over k , then $G = 0$*

Proof of lemma. Note that $G_{\bar{k}}$ is a constant group scheme over $\bar{k}$ and hence a product of constant group schemes of the form $\mathbb{Z}/\ell^m\mathbb{Z}$ where ℓ 's are prime. Then $G_{\bar{k}}^\vee$ is a product of μ_{ℓ^m} . Since μ_{p^m} is not reduced and we assume that $G^\vee$ is étale, p cannot be one of these ℓ . But this implies that the order of $G^\vee$ (and hence G) is prime to p . Hence $G_{\bar{k}} = 0$ since G is assumed to be p -power order. Because fields extensions are faithfully flat, we have $G = 0$ as well. $\square$

To construct the Dieudonné module associated to a finite commutative group schemes of p -power order, let W_n be the ring scheme of Witt vectors of length n , i.e. $W_n(k) = W(k)/p^n W(k)$. One can show that if $G^\vee$ is connected, then $\varinjlim_n \text{Hom}_k(G, W_n)$ is a Dieudonné module over k with endomorphisms F, V induced by $F_{G/S}, V_{G/S}$ respectively, and we set $M(G) = \varinjlim_n \text{Hom}_k(G, W_n)$. If $G^\vee$ is étale, by lemma 5.2 G itself is connected and we set $M(G) = M(G^\vee)^\vee$. Now use the multiplicative Jordan decomposition $G \cong G^{mu} \times G^{un}$ as in remark 5.7 which satisfies $(G^{mu})^\vee$ is étale and $(G^{un})^\vee$ is connected. Together they give $M(G) = M(G^{mu}) \oplus M(G^{un})$.

Example 5. For $\mathbb{Z}/p\mathbb{Z}$ and $\mathbb{Q}_p/\mathbb{Z}_p$, $M(\mathbb{Z}/p\mathbb{Z}) \cong k$ with $F = \sigma$ and $V = 0$ and $M(\mathbb{Q}_p/\mathbb{Z}_p) \cong W(k)$ with $F = \sigma$ and $V = p\sigma^{-1}$

We can further consider the "rationalization" of Dieudonné modules in $W(k)[1/p]$, namely **isocrystals**, which are the central objects in p -adic Hodge theory.

6 Lifting Problems of Abelian Varieties

6.1 Formal completions

First we restate smoothness and étaleness with their infinitesimal lifting property.

Definition 6.1. A morphism $f : X \rightarrow S$ is **formally smooth** (resp. **formally étale**) if for all affine S -schemes Y and all closed subschemes Y_0 of Y defined by nilpotent sheaves of ideals, the canonical map $\text{Hom}_S(Y, X) \rightarrow \text{Hom}_S(Y_0, X)$ is surjective (resp. bijective).

Remark 6.1. One can show that the assumption of Y being affine may be dropped by gluing morphisms.

Lemma 6.1. A morphism of schemes is smooth (resp. étale) if and only if it is locally of finite presentation and formally smooth (resp. formally étale).

Proof. See [BLR90, proposition 2.2.6] $\square$

In general, a **nilpotent thickening** of S is a closed immersion $X \rightarrow S$ whose defining sheaf of ideal $\mathcal{I}$ is nilpotent. We say it is a **k-thickening** if $\mathcal{I}^{k+1} = 0$

Recall that in Zariski topology, the **formal completion** of a scheme X along a closed subscheme Y defined by the sheaf of ideals $\mathcal{I}$ is $(\hat{X}, \mathcal{O}_{\hat{X}})$ where the underlying topological space of $\hat{X}$ is just Y and $\mathcal{O}_{\hat{X}} = \varprojlim \mathcal{O}_X/\mathcal{I}^n$. We now re-interpret them as sheave on the fppf site.

Definition 6.2. Let X be an abelian fppf sheaf on a scheme S and Y a subsheaf. Let $k \geq 1$. We define the **k-th infinitesimal neighborhood** of Y in X $\text{Inf}_Y^k(X) \subset X$ to be the subsheaf whose sections over T (an S -scheme) are those sections $t \in X(T)$ for which there is an fppf covering $\{T_i \rightarrow T\}$ and for each T_i a k -thickening $T'_i \rightarrow T_i$ such that $t|_{T'_i} \in Y(T'_i)$.

Example 6. If X is a S -scheme and Y is a closed subscheme defined by $\mathcal{I}$. Then one can show that $\text{Inf}_Y^k(X) \cong \text{Spec}(\mathcal{O}_X/\mathcal{I}^{k+1})$

Definition 6.3. The **formal completion** of X along Y is the subsheaf $\text{Inf}_Y(X) = \varinjlim \text{Inf}^k(G)$.

If X is a separated group scheme, which represents an abelian fppf sheaf of groups on S , then the unit section $e : S \rightarrow X$ is a closed immersion (this is clear since the diagonal map comes from the unit section by base change to X). We define the **formal completion** $\widehat{X}$ of X as $\text{Inf}_S(X)$ via the unit section. It is clear that $\widehat{X}$ is a subgroup of X . Moreover, example 6 implies that it is represented by the formal completion of X via the unit section in Zariski topology which is a formal scheme. In this case, this formal scheme will inherit the group structure from X and becomes a formal group.

If X is also smooth (e.g. an abelian scheme), there is more to tell about this formal group. If $S = \text{Spec}(R)$ is affine, by smoothness the completed local ring at the unity section $\widehat{\mathcal{O}}_{X,e}$ is known to be isomorphic with the formal power series $R[[x_1, \dots, x_n]]$ where n is the relative dimension at e . Hence the formal completion $\widehat{X}$ (as an fppf sheaf of groups) is represented by $\text{Spf}(R[[x_1, \dots, x_n]])$. In general, we can find an affine covering of S , such that over each affine chart $\text{Spec}(R_i)$ $\widehat{X}$ is represented by $\text{Spf}(R_i[[x_1, \dots, x_n]])$. This leads to the following definition.

Definition 6.4. A **formal Lie group** over R is a formal scheme isomorphic to $\text{Spf}(R[[x_1, \dots, x_n]])$ for some n .

Our preceding argument shows that for an abelian scheme A over S , $\widehat{A}$ is locally represented by a formal Lie group. Now given a formal Lie group $\text{Spf}(R[[x_1, \dots, x_n]])$ over a complete noetherian local ring R , the group structure gives the co-multiplication $m^* : R[[x_1, \dots, x_n]] \rightarrow R[[t_1, \dots, t_n, s_1, \dots, s_n]]$, which is continuous w.r.t. the maximal-adic topology. Then we define

$$F = (m^*(x_1), \dots, m^*(x_n))$$

Using to the continuity and co-associativity of m^* , one can show that F is a formal group law.

We now give its connection with p -divisible group. Naturally a p -divisible group over S represents an fppf sheaf of groups on S such that $[p]$ is fppf surjective (i.e. p -divisible). One can show that the converse is true provided that the p -torsion of the fppf sheaf of groups is represented by a finite flat group scheme. We can recover the whole structure using the short exact sequence in remark 5.3.

Messing showed that under certain conditions, the formal completion of a p -divisible group of groups is indeed a formal Lie group:

Theorem 6.1. (Grothendieck-Messing) Suppose that p is locally nilpotent on S and that G is a p -divisible group over S . Then G is formally smooth and $\widehat{G}$ is locally represented by a formal Lie group.

Proof. See [Mes72, chapter II] □

For the reason of this theorem, we will only consider the p -divisible group of abelian schemes over R where p is nilpotent, e.g. an artinian local ring R with residue field of characteristic p .

6.2 Deformations of abelian varieties and the Serre-Tate theorem

Definition 6.5. Let A be an abelian variety over a field k . For a complete local noetherian ring R with residue field k , a **deformation** of A over R is a pair $(\mathcal{A}, i)$ consisting of an abelian scheme $\mathcal{A}$ over R and an isomorphism $i : \mathcal{A}_k \simeq A$ over k . We use $\text{Def}(A/R)$ to denote the category of isomorphism classes of deformations of A over R .

We can give a similar definition for the deformation of p -divisible groups. Recall that we always use $A[p^\infty]$ to denote the p -divisible group of an abelian scheme A .

Theorem 6.2. (Serre-Tate) Let R be a ring in which a prime p is nilpotent, I a nilpotent ideal in R , and $R_0 = R/I$. For an abelian scheme A and p -divisible group G over R , let $A_0 = A \times_R R_0$ and $G_0 = G \times_R R_0$ denote their reductions over R_0 respectively.

For any abelian scheme A over R , let $\epsilon_A : A[p^\infty]_0 \simeq A_0[p^\infty]$ be the canonical isomorphism. Then the functor $A \rightsquigarrow (A_0, A[p^\infty], \epsilon_A)$ from the category of abelian schemes over R to the category of triples $(A', G, \epsilon : A'[p^\infty] \simeq G_0)$ where A' is an abelian scheme over R_0 is an equivalence of categories.

Corollary 6.1. Let R an artinian local ring with residue field k and A_0 be an abelian variety over k . The functor $\text{Def}(A_0/R) \rightarrow \text{Def}(A_0[p^\infty]/R)$ by sending the isomorphism class of the pair $(\mathcal{A}, i : \mathcal{A}_k \simeq A_0)$ to the isomorphism class of the pair $(\mathcal{A}[p^\infty], i : \mathcal{A}_k[p^\infty] \simeq A_0[p^\infty])$ is an equivalence of categories.

Remark 6.2. Corollary 6.1 shows that over artinian local rings, deforming an abelian variety is the same as deforming its p -divisible group. One can generalize theorem 6.2 by replacing R and R_0 with any nilpotent thickening $S_0 \rightarrow S$. Of course, we need to require p to be locally nilpotent on S as in Grothendieck-Messing.

For the rest of this section, we will give a sketch of Drinfeld's proof of the Serre-Tate theorem. Suppose ring R is N -torsion for some integer N and $I \triangleleft R$ is an ideal with $I^{v+1} = 0$ for some $v \geq 1$. For abelian fppf sheaves $G : \text{Alg}_R \rightarrow \text{Grp}$ and H on R and $f : G \rightarrow H$ an fppf homomorphism, let G_0, H_0 and f_0 denote their restrictions to Alg_{R_0} respectively. For every R -algebra A , define an fppf sheaves G_I by $G_I(A) = \ker(G(A) \rightarrow G(A/IA))$.

Lemma 6.2. Suppose G is a formal Lie group over R . Then $N^v G_I = 0$

Proof. Recall that G is of the form $\text{Spf} R[[x_1, \dots, x_n]]$ with its formal group law F . For any R -algebra A , its A -points are therefore given by $\underline{a} = (a_1, \dots, a_n)$ with $a_i \in A$ with addition $\underline{x} +_F \underline{y} = F(\underline{x}, \underline{y})$. If $\underline{a} \in G_I(A)$, then $a_i \in IA$ for all i .

Now by the axioms for the formal group law and induction, one can easily see that

$$[N]\underline{a} = N\underline{a} + \text{degree} \geq 2 \text{ terms of } \underline{a} = \text{degree} \geq 2 \text{ terms of } \underline{a} \in G_{I^2}(A) \quad (6.1)$$

since R and hence A are N -torsion. So $[N]G_I \subseteq G_{I^2}$. Do this v times, we get $[N^v]G_I \subseteq G_{I^{2^v}} \subseteq G_{I^{v+1}} = 0$ since $I^{v+1} = 0$. $\square$

Corollary 6.2. Suppose G is a fppf sheaves of groups such that $\widehat{G}$ is locally represented by a formal Lie group. Then $N^v G_I = 0$

Proof. Note that $G_I(A)$ consists of those sections of $G(A)$ that become the unit when restricted to the v -nilpotent thickening $\text{Spec}(A/IA) \rightarrow \text{Spec}(A)$. Then by definition $G_I \subset \text{Inf}^v(G) \subset \widehat{G}$, and hence $G_I = (\widehat{G})_I = G_I \cap \widehat{G}$. $\square$

Theorem 6.3. (Drinfeld's rigidity theorem) Let G and H be two abelian fppf sheaves on R . Suppose G is N -divisible, H is formally smooth and $\widehat{H}$ is locally represented by a formal Lie group. Then

(i) $\text{Hom}_R(G, H)$ and $\text{Hom}_{R/I}(G_0, H_0)$ have no N -torsion.

(ii) The restriction map $\text{Hom}_R(G, H) \rightarrow \text{Hom}_{R/I}(G_0, H_0)$ is injective.

Proof. (i) Since G is N -divisible, $[N] : G \rightarrow G$ is fppf surjective. Now any $f \in \text{Hom}_R(G, H)$ commutes with $[N]$. Hence $[N]f = f[N] = 0$ implies $f = 0$. Note that its restriction G_0 is N -divisible as well.

(ii) Suppose f is in the kernel of this map. Functoriality gives the following commutative diagram

$$\begin{array}{ccc} G(A) & \longrightarrow & G(A/IA) \\ \downarrow f(A) & & \downarrow 0 \\ H(A) & \longrightarrow & H(A/IA) \end{array} \quad (6.2)$$

which shows that f lands in H_I . By corollary 6.2, $N^v H_I = 0$ and hence $[N^v]f = 0$. Now use part (i) to get $f = 0$ $\square$

This observation by Drinfeld is the heart in the proof of the Serre-Tate and the following lemmas regarding the lifting of homomorphisms.

Proposition 6.1. *With the settings in theorem 6.3, for any $f_0 : G_0 \rightarrow H_0$, there exists a unique map $f_N : G \rightarrow H$ lifting $N^v f_0$*

Proof. Theorem 6.3 shows that if such a lift exists then it is unique. Now define $f_N(A)$ for any R -algebra A as the composition

$$G(A) \rightarrow G(A/IA) \rightarrow H(A/IA) \rightarrow H(A) \quad (6.3)$$

where the middle map is $f_0(A/IA)$ as A/IA is an R_0 -algebra and the right map is $[N^v]s$, where s is any set-theoretical section of $H(A) \rightarrow H(A/IA)$ which is surjective since H is formally smooth. The resulting map does not depend on the choice of s . Indeed, if s and s' are two sections of the reduction map $h : H(A) \rightarrow H(A/IA)$, then for any $x \in H(A/IA)$, $h(s - s')(x) = 0$ and hence $s - s'$ lands in $H_I(A)$. By corollary 6.2, $[N^v]s = [N^v]s'$. Functoriality of f_N can be shown similarly. By our construction, f_N is a fppf homomorphism that lifts $N^v f_0$ since $IA = 0$ when A is an R_0 -algebra. $\square$

Proposition 6.2. *$f_0 : G_0 \rightarrow H_0$ can be lifted to a homomorphism $f' : G \rightarrow H$ if and only if the map f_N that lifts $N^v f_0$ annihilates $G[N^v]$*

Proof. If f lifts f_0 , then $N^v f$ lifts $N^v f_0$. By theorem 6.3, $f_N = N^v f$ and hence annihilates $G[N^v]$. Conversely, suppose f_N kills $G[N^v]$ and hence it factors through $G/G[N^v]$. By the N -divisibility of G there is a short exact sequence

$$0 \rightarrow G[N^v] \rightarrow G \xrightarrow{[N^v]} G \rightarrow 0$$

which gives an isomorphism $s : G/G[N^v] \xrightarrow{[N^v]} G$. Then $f_N \circ s^{-1} = f' : G \rightarrow H$ is an fppf homomorphism such that $f_N = N^v f'$. Let f'_0 be its restriction on Alg_{R_0} . Then $N^v f'_0 = (N^v f')_0 = N^v f_0$ shows $f'_0 = f_0$ since part (i) of theorem 6.6 shows that $\text{Hom}_{R/I}(G_0, H_0)$ has no N -torsion, which implies that f' lifts f_0 $\square$

Now we begin the proof of Serre-Tate.

Proof of fully faithfulness. Given two abelian schemes A and B over R where p is locally nilpotent and morphisms $f[p^\infty] : A[p^\infty] \rightarrow B[p^\infty]$, $f_0 : A_0 \rightarrow B_0$ such that $f_0[p^\infty] = (f[p^\infty])_0$, we want to show there is a unique $f' : A \rightarrow B$ that induces all of them. Note that A is p -divisible since $[p]$ is fppf surjective on the abelian scheme A . By theorem 6.1, A is formally smooth and $\hat{A}$ is a formal Lie group. Hence we apply part (ii) of theorem 6.3 which shows that if such f' exists then it is unique.

By proposition 6.1 we have $f_p : A \rightarrow B$ that lifts $p^v f_0$. Hence $f_p[p^\infty]$ lifts $p^v f_0[p^\infty] = (p^v f[p^\infty])_0$ which is also lifted by $p^v f[p^\infty]$. Since p -divisible groups are p -divisible as fppf sheaves, we can apply part (ii) of theorem 6.3 again and get $f_p[p^\infty] = p^v f[p^\infty]$. It follows that $f_p[p^\infty]$ (and hence f_p) kills $A[p^v]$. By proposition 6.2, f_0 can be lifted to $f' : A \rightarrow B$ and hence $p^v f' = f_p$. Then $p^v f'[p^\infty] = f_p[p^\infty] = p^v f[p^\infty]$. By part (i) of theorem 6.3, $f'[p^\infty] = f[p^\infty]$. So f' induces both f_0 and $f[p^\infty]$ $\square$

To prove essential surjectivity, we need the following wonderful theorem by Grothendieck

Theorem 6.4. *Let $S_0 \rightarrow S$ be a nilpotent thickening and A_0 an abelian scheme over S_0 . Then A_0 lifts to an abelian scheme over S .*

This theorem says that the **obstruction** to lift an abelian scheme A_0/S_0 to S vanishes. However, the resulting lifts do not always come in a natural way, e.g. the p -divisible group of the lift may not be a lift of the p -divisible group of the original abelian scheme. The Serre-Tate theorem essentially solves this problem.

Proof of essential surjectivity. We are given a triple $(A_0, G, \epsilon : G_0 \simeq A_0[p^\infty])$. By theorem 6.4, we can take an abelian scheme A/R that lifts A_0/R_0 . This yields an isomorphism $i : (A[p^\infty])_0 \simeq A_0[p^\infty] \simeq G_0$. By proposition 6.1 we take $i_p : A[p^\infty] \rightarrow G$ to be the canonical lift of $p^v i$ and $j_p : G \rightarrow A[p^\infty]$ to be the canonical lift of $p^v i^{-1}$. Then $i_p \circ j_p$ and $j_p \circ i_p$ lift $[p^{2v}]$ on G_0 and on $A_0[p^\infty]$ respectively. By theorem 6.3, $i_p \circ j_p = [p^{2v}]$ on G and $j_p \circ i_p = [p^{2v}]$ on $A[p^\infty]$ (clearly $[p^{2v}]_G$ also lifts $[p^{2v}]_{G_0}$) which are both fppf surjective. Hence i_p is an isogeny as its kernel $K \subseteq A[p^{2v}]$ is finite.

We want to show K is flat over R . By base-change, it is enough to prove that i_p is flat. By the fibrewise criterion of flatness and the fact that any fibre of the morphism factor through $\text{Spec} R_0$, it suffices to show that $(i_p)_0 = p^v i$ is flat, which is clear since i is an isomorphism.

Now the results on the quotient of group schemes show that the quotient A/K is an abelian scheme over R . Moreover, $(A/K)[p^\infty] = A[p^\infty]/K \cong G$ and $(A/K)_0 = A_0/A_0[p^v] \xrightarrow{p^v} A_0$ is an isomorphism since i_p lifts $p^v i$ and hence $K = \ker(i_p)$ lifts $\ker(p^v i) = A_0[p^v]$, which shows that A/K satisfies all desired properties. This completes Drinfeld's proof of the Serre-Tate theorem. $\square$

Remark 6.3. There is another Serre-Tate equivalence between the categories of divisible formal Lie groups and connected p -divisible groups over R , which can be found in [Tat67].

Serre-Tate theorem ensures that when lifting abelian varieties we can lift their associated p -divisible groups instead. The p -divisible groups are generally easier to handle, not only because they correspond to Dieudonné modules which are semi-linear objects. Their isogeny classes and endomorphism algebras are also easier to understand than abelian varieties. See [CCO14, theorem 3.1.3].

6.3 Canonical Lifts for Ordinary Abelian Varieties

One application of the Serre-Tate theorem is to find the **canonical lifts** for abelian varieties over k . To fulfill the condition for Grothendieck-Messing, we need to lift them to some rings where p is nilpotent. A natural candidate would then be $W_n(k) = W(k)/p^n W(k)$, the ring of Witt vectors of length n .

Note that each projection $W_n(k) \rightarrow W_{n-1}(k)$ gives a nilpotent thickening. Therefore, starting with an abelian variety $A = A_1$ over $k = W_1(k)$, we can produce a sequence of abelian schemes A_n over $W_n(k)$ such that $A_n \times_{W_n(k)} W_m(k) = A_m$ for any $m \leq n$ using theorem 6.4 repetitively, which then induces natural projections $p_{m,n} : A_m \rightarrow A_n$. So together they form an inductive system and the colimit $\tilde{A}$ is in general a formal abelian scheme over $W(k)$. To make $\tilde{A}$ algebraic, we can use Grothendieck's algebraicity theorem by adding more restrictions on A . But even so, we barely know anything about $\tilde{A}$ other than it exists.

We are more interested when A is ordinary. Recall that an abelian variety A over a perfect field k with characteristic p of dimension g is **ordinary** if $A[p](\bar{k}) \cong (\mathbb{Z}/p\mathbb{Z})^g$. In general we have $A[p](\bar{k}) \cong (\mathbb{Z}/p\mathbb{Z})^{r_A}$ where $0 \leq r_A \leq g$, and r_A is defined to be the **p-rank** of A .

To understand this, recall that in section 5.3 we have the decomposition $A[p] = A[p]_{ll} \times A[p]_{le} \times A[p]_{el}$. Since $[p]_A$ is the dual homomorphism of $[p]_{A^\vee}$, theorem 1.6 gives $A^\vee[p] = A[p]^\vee = A[p]_{ll}^\vee \times A[p]_{le}^\vee \times A[p]_{el}^\vee$ and so $A^\vee[p]_{ll} = A[p]_{ll}^\vee$, $A[p]_{le}^\vee = A^\vee[p]_{el}$ and $A[p]_{el}^\vee = A^\vee[p]_{le}$. Note that $(A[p]_{el})_{\bar{k}}$ is a p -torsion constant group scheme and so has the form of $(\mathbb{Z}/p\mathbb{Z})^{r_A}$ where r_A is exactly the p -rank of A . Use the short exact sequence in remark 5.3 we see that $(A[p^k]_{el})_{\bar{k}} = (\mathbb{Z}/p^k\mathbb{Z})^{r_A}$ for any $k \geq 1$.

Lemma 6.3. *The p -rank is isogeny-invariant, i.e. $r_A = r_{A'}$ for isogenous A and A' .*

Proof. Let $f : A \rightarrow A'$ be an isogeny. Then it induces homomorphisms $A[p^k] \rightarrow A'[p^k]$ for each k . The observations above give $p^{kr_A} = \#|A[p^k](\bar{k})| \leq \#|ker(f)(\bar{k})| \#|A'[p^k](\bar{k})| = \#|ker(f)(\bar{k})| \cdot p^{kr_{A'}}$. Passing k to the infinity, we get $r_A \leq r_{A'}$. Take the dual isogeny and we see $r_{A'} \leq r_A$ also. $\square$

So $A^\vee[p]_{el}$ is of the order p^{r_A} . Since $A[p]_{le}^\vee = A^\vee[p]_{el}$, this implies that $A[p]_{le}$ has the order p^{r_A} as well. Now it follows from the decomposition of $A[p]$ that $r_A \leq g$ and A is ordinary if and only if the local-local part $A[p]_{ll}$ vanishes.

Indeed, the local-local part is exactly the obstruction for the canonical lifting because étale schemes can be lifted canonically through nilpotent thickenings:

Proposition 6.3. *Let $S_0 \rightarrow S$ be a nilpotent thickening. There is an equivalence between categories of étale schemes over S and S_0 by the functor $X \rightsquigarrow X_0 = X \times_S S_0$. Moreover, given X_0/S_0 , the étale scheme X/S that lifts X_0 is unique up to isomorphism.*

This can be seen from the infinitesimal lifting property. As a result, $A[p]_{el}$ admits a unique lift to any artinian local ring R with residue field k . We denote it by $\tilde{A}[p]^{et}$. For $A[p]_{le}$, it also has a canonical lift by taking the Cartier dual of the unique lift of its étale dual. We denote it by $\tilde{A}[p]^o$.

From now on we assume A to be ordinary. Passing to the colimits, we see that the étale part and connected part of the p -divisible group $A[p^\infty]$ have a canonical lift respectively. By Serre-Tate, lifting A to R is the same as to lift $A[p^\infty]$ to R , which then amounts to finding an extension of those two canonical lifts. Taking the p -torsion, this results in an extension $0 \rightarrow \tilde{A}[p]^o \rightarrow E \rightarrow \tilde{A}[p]^{et} \rightarrow 0$ on level-1. On the other hand, any such extension can recover a lift by remark 5.3.

So we get a correspondence from the deformations of A to R to $Ext^1(\tilde{A}[p]^o, \tilde{A}[p]^{et})$, which is in fact an equivalence of categories. The lift mapped to the 0 element of the latter group is our **canonical lift** of A to R .

If k is algebraically closed (e.g. $\overline{\mathbb{F}_p}$), then we know that $A[p]^{et} = (\mathbb{Z}/p\mathbb{Z})^g$ and $A[p]^o = \mu_p^g$ instead of just being their twists. We pass to the colimits and consider the extensions of p -divisible groups $Ext^1(\mu_{p^\infty}^g, (\mathbb{Q}_p/\mathbb{Z}_p)^g)$. First we need to make some adjustment to the notations. Note that the p -adic Tate module $T_p A(k) \cong \mathbb{Z}_p^g$ since A is ordinary, so naturally $T_p A(k) \otimes_{\mathbb{Z}_p} \mathbb{Q}_p/\mathbb{Z}_p = (\mathbb{Q}_p/\mathbb{Z}_p)^g$.

Moreover, earlier observation shows that $\mu_{p^k}^g = A[p^k]^o = (A^\vee[p^k]^{et})^\vee = (A^\vee[p^k](k))^\vee = Hom(A^\vee[p^k](k), \mathbb{G}_m)$ by the definition of Cartier duality. Then $\mu_{p^k}^g(R) = Hom_{\mathbb{Z}}(A^\vee[p^k](k), \mathbb{G}_m(R)) = Hom_{\mathbb{Z}}(A^\vee[p^k](k), \mu_{p^k}(R))$. Passing to colimits, we see that $A[p^\infty]^o(R) = \mu_{p^\infty}^g(R) = Hom_{\mathbb{Z}_p}(T_p A^\vee(k), \mu_{p^\infty}(R)) = Hom_{\mathbb{Z}_p}(T_p A^\vee(k), \widehat{\mathbb{G}_m}(R))$.

These are just preparations to associate the deformations of A to R with linear objects. Now let $\hat{A}$ be the formal completion of A along the unit section.

Lemma 6.4. *With the preceding settings, $\hat{A} \cong A[p^\infty]^o$ as fppf sheaves on R .*

Proof. Note that $A[p^\infty]^o = \varinjlim A[p^n]^o$. Each $A[p^n]^o$ is connected and so its augmentation ideal is nilpotent. Hence $A[p^n]^o \hookrightarrow A$ factors through some infinitesimal neighborhood of identity $Inf^k(X)$. So $A[p^\infty]^o \subset \hat{A}$. On the other hand, the formal group law on the formal Lie group $\hat{A}$ as shown in lemma 6.2 implies that $\hat{A}$ is killed by some power of p as p is nilpotent in R . So the converse is also true. $\square$

Consider the natural short exact sequence $0 \rightarrow T_p A(k) \rightarrow T_p A(k) \otimes_{\mathbb{Z}_p} \mathbb{Q}_p \rightarrow T_p A(k) \otimes_{\mathbb{Z}_p} \mathbb{Q}_p/\mathbb{Z}_p \rightarrow 0$. The functor $Hom(-, \hat{A})$ induces a long exact sequence

$$\dots \rightarrow Hom(T_p A(k) \otimes_{\mathbb{Z}_p} \mathbb{Q}_p, \hat{A}) \rightarrow Hom(T_p A(k), \hat{A}) \xrightarrow{\delta} Ext^1(T_p A(k) \otimes_{\mathbb{Z}_p} \mathbb{Q}_p/\mathbb{Z}_p, \hat{A}) \rightarrow Ext^1(T_p A(k) \otimes_{\mathbb{Z}_p} \mathbb{Q}_p, \hat{A}) \rightarrow \dots$$

Theorem 6.5. *δ is an isomorphism.*

Note that $T_p A(k) \otimes_{\mathbb{Z}_p} \mathbb{Q}_p$ is p -divisible since it is a $\mathbb{Q}_p$ -module, while $\hat{A}$ is killed by some power of p as noted in lemma 6.4. So $\text{Hom}(T_p A(k) \otimes_{\mathbb{Z}_p} \mathbb{Q}_p, \hat{A}) = 0$ and δ is injective.

To show surjectivity, we can instead show the map $\text{Ext}^1(T_p A(k) \otimes_{\mathbb{Z}_p} \mathbb{Q}_p, \hat{A}) \rightarrow \text{Ext}^1(T_p A(k), \hat{A})$ in the long exact sequence is injective. As $T_p A(k) \cong \mathbb{Z}_p^g$ and $\hat{A} \cong A[p^\infty]^\circ \cong \mu_{p^\infty}^g$, it suffices to show that $\text{Ext}^1(\mathbb{Q}_p, \mu_{p^\infty}) \rightarrow \text{Ext}^1(\mathbb{Z}_p, \mu_{p^\infty})$ is injective. This is the most difficult part that uses spectral sequences from homological algebra, and we will have to omit its proof.

Definition 6.6. For an abelian variety A over k , the **local moduli functor** $\widehat{\mathcal{M}}_{A/k} : \text{Art}_k \rightarrow \text{Set}$ assigns to every artinian local k -algebra R with residue field k the set of isomorphism classes of deformations of A over R , i.e. $R \rightsquigarrow \text{Def}(A/R)$.

Combining everything in this section so far, we get the following theorem

Theorem 6.6. For an ordinary abelian variety A over an algebraically closed field k with characteristic p , there is a natural isomorphism

$$\widehat{\mathcal{M}}_{A/k}(R) \xrightarrow{\sim} \text{Hom}_{\mathbb{Z}_p}(T_p A(k), \text{Hom}_{\mathbb{Z}_p}(T_p A^\vee(k), \widehat{\mathbb{G}}_m(R))) \simeq \text{Hom}_{\mathbb{Z}_p}(T_p A(k) \otimes_{\mathbb{Z}_p} T_p A^\vee(k), \widehat{\mathbb{G}}_m(R))$$

Remark 6.4. Theorem 6.6 says that the deformations of A over R can be identified with the $\mathbb{Z}_p$ -bilinear maps $q : T_p A(k) \times T_p A^\vee(k) \rightarrow \widehat{\mathbb{G}}_m(R) = 1 + m_R$. The canonical lift of A is the one that corresponds to the trivial bilinear map.

Since $T_p A(k)$ is a free $\mathbb{Z}_p$ module of rank g as A is ordinary, this is determined by the images of the g^2 free basis elements in $1 + m_R$, which are called the canonical coordinates in [Kat81]

Moreover, let A_n^{can} be the canonical lift of A to $W_n(k)$. As in the beginning of this section, they together with the projections form an inductive system. Messing in [Mes72, chapter V theorem 3.3] showed that the colimit in this case is an abelian scheme over $W(k)$, which is the canonical lift A^{can} of A to $W(k)$. Every homomorphism $f : A \rightarrow B$ uniquely lifts to an $f' : A^{\text{can}} \rightarrow B^{\text{can}}$. See [Kat81, theorem 2.1].

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